

Core Model Proposal #404:

Base Year Update: Move Model Base Year to 2021

Product: Global Change Analysis Model (GCAM)

Institution: Joint Global Change Research Institute (JGCRI)

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Related sector: all

Type of development: gcamdata

Purpose: The purpose of this core model proposal is to update key datasets and move the final calibration year to 2021.

Purpose

The purpose of this core model proposal is to update GCAM's base year to 2021. This requires updating key datasets and making additional data processing improvements to advance the final model calibration year to 2021. The data processing improvements include: 1) Add code to warn then extend historical datasets by copying forward to cover the extended historical time period 2) Implement and utilize a way to use the "set_data" model year constants in our assumption files that get processed via gather_years 3) Other minor issues such as removing assumptions around exact 5-year or otherwise constant time steps.

Description of Changes

Update Key Datasets

IEA Energy Balance

The IEA Energy Balances Database is one of the key input data sets to GCAM and provides information on historical energy flows. In this proposal, we update GCAM to use a newer version of the IEA Energy Balances Database (Year 2023) which has values through the Year 2022). IEA changed the "Total Primary Energy Supply" Flow constant from "TPES" to "TES". The former was hard coded in several chunks thus we went ahead and changed it to a constant so it can easily be updated if it changes again in the future. Note that the IEA Energy Balances Database is a proprietary data set and not released with GCAM; however, some regionally and sectorally aggregated derivative products of the IEA Energy Balances Database are included in gcam as binary "prebuilt data" so that community users can run GCAM even if they do not have a license for the IEA Energy Balances Database.

energy/comtrade_ff_trade

Using a simple methodology to extend primary fossil fuel trade proved insufficient thus we go ahead and update this data source form UN COMTRADE.

AgLU and FAO Data

All FAO data needed for GCAM is generated using an updated version of gcamfaostat (see the v1.1.0-beta release version; <https://github.com/JGCRI/gcamfaostat/releases/tag/v1.1.0-beta>; the raw data is archived so the process is fully traceable and replicable). In addition to FAO data, other related AgLU data and gcamdata modules are updated, as summarized by Table 1 and Table 2, below.

Table 1. AgLU data updates

Data	File Location	Old Name (if changed)	Description
GCAMFAOSTAT_ metadata	<i>aglu/FAO</i>		A summary of data generated by <i>gcamfaostat</i> and used in <i>gcamdata</i> .
GCAMFAOSTAT_ BiTrade	<i>aglu/FAO</i>	GCAMDATA_FAOSTAT_ BiTrade_194Regs_400Items_20 10to2020	Year: 2010 - 2022; 194 countries; 533 items.
GCAMFAOSTAT_ SUA	<i>aglu/FAO</i>	GCAMDATA_FAOSTAT_ SUA_195Regs_530Items_2010t o2019	Year: 2010 - 2022; 195 countries; 529 items.
GCAMFAOSTAT_ FBSH_CB	<i>aglu/FAO</i>	GCAMDATA_FAOSTAT_ FBSH_CB_ 173Regs_118Items_1973to2009	Year: 1973-2022; 173 countries; 118items.
GCAMFAOSTAT_ NonFodderProdArea	<i>aglu/FAO</i>	GCAMDATA_FAOSTAT_ ProdArea_ 195Regs_271Prod160AreaItems _1973to2020	Year: 1973-2022; 195 countries; 272 items.
GCAMFAOSTAT_ FodderProdArea	<i>aglu/FAO</i>	GCAMDATA_FAOSTAT_ ProdArea_96Regs_ 16FodderItems_1973to2020	Year: 1973-2022; 96 countries; 16 items.
GCAMFAOSTAT_ AnimalStock	<i>aglu/FAO</i>	GCAMDATA_FAOSTAT_ AnimalStock_202Regs_ 22Items_1973to2020	Year: 1973-2022; 202 countries; 22 items.
GCAMFAOSTAT_ ProdPrice	<i>aglu/FAO</i>	GCAMDATA_FAOSTAT_ ProducerPrice_ 170Regs_185PrimaryItems_ 2010to2020	Year: 1973-2022; 172 countries; 180 items.
GCAMFAOSTAT_ MacroNutrientRate	<i>aglu/FAO</i>	GCAMDATA_FAOSTAT_ MacroNutrientRate_ 179Regs_426Items_ 2010to2019Mean	Year: 1973-2022; 188 countries; 432 items.
GCAMFAOSTAT_ ForProdTrade	<i>aglu/FAO</i>	FAO_Paper_Prod_t_FORESTA T and GCAMDATA_FAOSTAT_ ForProdTrade_ 215Regs_Roundwood_1973to20 20	Year: 1973-2022; 185 countries; 6 items.
GCAMFAOSTAT_ ForExpPrice	<i>aglu/FAO</i>	GCAMDATA_FAOSTAT_ ForExportPrice_ 214Regs_Roundwood_1973to20 20	Year: 1973-2022; 183 countries; 6 items.
GCAMFAOSTAT_ LandCover	<i>aglu/FAO</i>	GCAMDATA_FAOSTAT_ LandCover_229Regs_	Year: 1973-2022; 229 countries; 3 items.

		3Covers_1973to2020	
GCAMFAOSTAT_ NFertilizer	<i>aglu/FAO</i>	GCAMDATA_FAOSTAT_ NFertilizerProdDemand_ 175Regs_1Item_1973to2020	Year: 1973-2022; 212 countries; 1 items.
FAO_ag_items_ PRODSTAT	<i>aglu/FAO</i>		Mapping updated
FAO_an_items_ PRODSTAT	<i>aglu/FAO</i>		Mapping updated
Mapping_item_ FBS_GCAM	<i>aglu/FAO</i>		Mapping updated
Mapping_SUA_ PrimaryEquivalent	<i>aglu/FAO</i>		Mapping updated
SUA_item_code_map	<i>aglu/FAO</i>		Mapping updated
USDA_Cost_database	<i>aglu/USDA</i>		Removed as not used
USDA_Alalfa_prices_U SDt	<i>aglu/USDA</i>		Year: 1973-2023; USA; 1 items.
AGLU_etry	<i>aglu</i>		Mapping updated
GCAMFAOSTAT_ CapitalStock	socioeconomics		Year: 1973-2022; 186 countries; 3 items; data will be used in KLEAM.
GCAMFAOSTAT_GDP	socioeconomics /GDP		Year: 1973-2022; 212 countries; 1 items.
FAO_GDP_Deflators	common		Year: 1973-2022; 212 countries; 1 items.

Table 2, below, summarizes the main AgLU-related gcamdata chunk updates (excluding chunks with only minor changes).

Table 2. AgLU data processing updates

Chunk	Description
constants	<pre> aglu.MODEL_MEAN_PERIOD_LENGTH <- 3 aglu.MODEL_PRICE_YEARS <- 2020:2022 aglu.MODEL_MACRONUTRIENT_YEARS <- 2020:2022 aglu.FALLOW_YEARS <- 2020:2022 </pre>
module_aglu_L100.FAO_preprocessing_OtherData	FAO land data had item name changes. E.g., "Land with temporary fallow" is changed to "Temporary fallow". They are updated and assertions are added.
module_aglu_L100.FAO_SUA_PrimaryEquivalent	Improvements were added in PCe and data flows. Additional assertions were added.
module_aglu_L100.regional_ag_an_for_prices	Improvements in processing (e.g., deals with zero values).
module_aglu_L109.ag_an_ALL_R_C_Y	Improvements in data balancing and trade adjustments

For the AgLU data years, GCAM previously used a 5-year average centered around the base year. In this proposal, we have shifted to a 3-year average centered on 2021, the new base year. This change is primarily due to the fact that FAOSTAT has not yet released complete data for 2023, and using an asymmetric 5-year rolling average would result in a shifted mean/center year. Specifically, `aglu.MODEL_MEAN_PERIOD_LENGTH` is set to 3 and related year changes are made accordingly.

Base-year CEDS Emissions Data

CEDS historical emissions data for air pollutants, CH₄, and N₂O were updated to version v2024_07_08. Changes in the historical data in this version can be seen [here](#).

Extending datasets by copying them forward

There are several datasets used in `gcamdata` which have not yet been updated to ensure that they include a full historical time series up to and including 2021. The reasons for not updating these datasets are varied but often is the case because an update is not available. In each such case, we ensure that a warning is generated to remind the user that they should consider updating the data (or find an alternative), before the code extends the data. Because this operation occurs in several places, we created a couple of helper functions (`copy_data_forward_long` and `copy_data_forward_wide`). In some cases it was easier to extend the data at the start of the

processing; in cases it was easier to extend the data at the end of the processing. It should also be noted that it is not always best to extend by copying forward and an alternative method should be used which is detailed on a case by case basis below. Table 3 provides an overview of where data were extended.

Table 3. Places in gcamdata where data is extended to cover the historical time series up to and including 2021

Chunk	Data
module_emissions_L102.nonco2_ceds_R_S_Y	*L102.ceds_GFED_nonco2_tg_R_S_F
module_emissions_L102.nonco2_ceds_R_S_Y	*L102.ceds_GFED_nonco2_tg_C_S_F
module_emissions_L102.nonco2_ceds_R_S_Y	*L102.ceds_int_shipping_nonco2_tg_S_F
module_energy_L1323.iron_steel	energy/steel_prod_process
module_energy_L1323.iron_steel	energy/WSA_direct_reduced_iron_2008_2019
module_energy_L1326.aluminum	energy/aluminum_prod USGS or CEDS
module_energy_L1326.aluminum	energy/aluminum_prod_region_IAI
module_gcamusa_L101.EIA_SEDS	L101.EIA_SEDS

* Note: the capability exists to extend this data however, given the CEDS data was updated the series extension is not currently in effect.

Some additional changes related to this issue but not handled in the standard way (described above) include:

zenergy_L113.atb_cost

There are two year constants which govern behavior in this processing: energy.ATB_BASE_YEAR and energy.ATB_LATEST_YEAR and the two were used interchangeably in the chunk. However, the intent was the former was meant to be a property of the data while the later was to be a user assumption of when to switch from ATB costs to the standard A23 assumption files. The module_energy_L113.atb_cost chunk was updated accordingly. In addition if energy.ATB_LATEST_YEAR > energy.ATB_BASE_YEAR we warn and extend the ATB data extrapolating using just the ATB assumptions to include energy.ATB_LATEST_YEAR. And of course we set energy.ATB_LATEST_YEAR to now be the MODEL_FINAL_BASE_YEAR = 2021.

zgcamusa_L100.Socioeconomics

We updated the BEA GDP data to include values through 2023 and included a check to warn and error if the GCAM-USA historical dataset does not cover the full set of historical years.

zsocio_L100.Population_downscale_ctry

We updated the UN historical population data to extend to 2023 as the "base" SSP year is still 2020. Some related code changes to address changing ISOs were also required.

zsocio_L180.GDP_macro

We copy forward the Penn World table data to cover the new historical period. However, we could not use the standard methods (described above) as some NA values are expected in the data.

zenergy_L1327.paper.R

A supplementary data source was needed for paper processing to estimate China's historical biomass energy use in this sector. No updated data is available so we warn, then extend the data by using the relative growth in the paper sector as a whole.

zenergy_L1011.en_bal_adj.R

A supplementary data source was needed to adjust international shipping using EIA data, however that source is no longer being updated. Thus we warn, then extend the EIA data by using the relative growth in international shipping from the IEA data.

Avoid hard coding Calibration-to-Future years in assumption files

To help gcamdata handle different base year definitions more flexibly, we expand the capabilities of the `gather_years` function to allow users to use the same model time constants which were available in `set_years`.

Constant	Value
start-year	<code>min(MODEL_BASE_YEARS)</code>
final-calibration-year	<code>MODEL_FINAL_BASE_YEAR</code>
final-historical-year	<code>as.numeric(max(HISTORICAL_YEARS))</code>
initial-future-year	<code>min(MODEL_FUTURE_YEARS)</code>
initial-nonhistorical-year	<code>min(MODEL_YEARS[MODEL_YEARS > max(HISTORICAL_YEARS)])</code>
end-year	<code>max(MODEL_FUTURE_YEARS)</code>

Note, almost all of the gcamdata assumption files (i.e. A23.globaltech_shrwt.csv) use the gather_years function as the first step to produce the "standard" way of processing these files (gather_years → complete(year=MODEL_YEARS) → fill missing with approx_fun). This is important as quite often these files are used to specify on/off behavior for new technologies (e.g., when emerging technologies like CCS or H2 initially become available). When these on/off years are hard coded, we can end up with unintuitive issues (e.g., failure to calibrate historical periods or solve future ones due to a technology being on/off at the wrong time). Using the year constants as part of gather_years creates a robust system for navigating the final calibration year / initial future year boundary. While it does not by itself ensure that a future technology availability doesn't fall before the final calibration year, gather_years will perform strict error checking to ensure all resolved years are in order. This allows errors to be flagged at the source (when the assumption file is processed) instead of much later (e.g., as a calibration failure when running the model).

Only a subset of the assumption files were updated to utilize this new feature, specifically the ones which would generate errors when specifying a new model base year. The ones that have been updated include:

Assumption File
energy/A21.globalrsrtech_coef
energy/A21.globalrsrtech_cost
energy/A22.globaltech_shrwt
energy/A23.globaltech_capital_adv
energy/A23.globaltech_capital_low
energy/A23.globaltech_capital
energy/A23.globaltech_OMfixed
energy/A23.globaltech_OMvar
energy/A23.globaltech_shrwt
energy/A23.subsector_shrwt_coal_R
energy/A23.subsector_shrwt_nuc_R
energy/A25.globaltech_shrwt
energy/A25.subsector_shrwt
energy/A26.globaltech_shrwt
energy/A26.subsector_interp
energy/A44.globaltech_shrwt
energy/A54.globaltranTech_interp_revised

energy/A321.subsector_interp
energy/A322.subsector_interp
energy/A323.globaltech_shrwt
energy/A324.globaltech_shrwt
energy/A327.subsector_interp_adj_future_years
energy/A327.subsector_interp
energy/A328.subsector_interp
gcam-usa/A23.elecS_globalinttech_shrwt
gcam-usa/A23.elecS_globaltech_shrwt
gcam-usa/A23.elecS_subsector_shrwt_interpto

Other Changes

input/extra/renewable_shareweights_ssp1_32.xml

This file contains some tuning from the very original SSP assumptions which appear to be implemented / edited by hand (including assumptions about model time steps). We update values in the XML to be consistent with the new model time steps.

input/gcamdata/inst/extdata/energy/A10.ResSubresourceProdLifetime.csv

We shortened average production lifetimes for crude oil and unconventional oil (see table below). In GCAM's resource-reserve model, when a region expands resource production, a quantity of resource equal to new production * average production lifetime is moved from the resource curve and into reserves. We observed that the 60 year lifetime was likely pushing regions up their crude oil resource curves too quickly. [Industry](#) suggests that oil and gas average 20-40 years of significant production; we assume a 30 year average production lifetime. For consistency, we also update the average production lifetime for unconventional oil to be consistent with the new value for crude oil, since the two subresources directly compete within the crude oil nest.

resource	average production lifetime	
	old	new
crude oil	60	30
unconventional oil	40	30
natural gas	30	30
coal	40	40

zemissions_L241.fgas

The "future" HFC emissions data processing assumed that 2020 was the first future period. We ensure only model periods are ultimately included in this table. Note that the "future" HFC emissions coefficients actually include some values in the historical period to take care of a couple of regions with missing input emissions data so it is not appropriate to simply filter for MODEL_FUTURE_YEARS.

zemissions_L252.MACC

We fixed a filter to properly ensure that we only include the MACC in future model periods.

zenergy_L210.resources

This chunk contains checks to ensure that we have enough resource in our supply curves to always cover calibrated historical demands and add resource to exactly cover those demands if needed. However, for some regions, we still observed calibration failures in 2021. This was because we were rounding the available grades too coarsely. We update the rounding to use `energy.DIGITS_CALOUTPUT (7)` digits to avoid this.

There was also inappropriate processing of assumption files for global tech cost and coef. We fix this by just using the standard methodology (`gather_years → complete(year=MODEL_YEARS) → fill missing with approx_fun`).

zenergy_L222.en_transformation

As with **zenergy_L210.resources**, we identified inappropriate processing of assumption file `A22.globaltech_shrwt` and we reverted this to standard processing.

zenergy_L244.building_det

We corrected a couple of places that were assuming old timesteps or final model year.

zenergy_L1327.paper

We improved checks for very small but non-zero amounts of biomass input to produce wood pulp and reset such IO coefficients to zero to avoid calibration errors.

zgamusa_L143.HDDCDD

We made changes to ensure that year columns are in order; without this, we can encounter an error with the new `gather_years` functionality.

zgamusa_L2241.coal_retire

We moved the vintage bin definitions into this chunk (instead of constants.R) so that we can come up with them dynamically as the model historical years change.

**zgcamusa_L1233.elec_water / zgcamusa_L2233.elec_segments_water /
zgcamusa_L2236.elecS_ghg_emissions**

We updated the processing to use a cooling tech mapping generated in the L2241.coal_retire chunk to ensure we will have the correct coal vintage bins.

zsocio_L280.GDP_macro

We cleaned up some assumptions of constant time steps when doing a moving average of labor productivity.

zgcamusa_L2247.elecS_tech_costs / zgcamusa_L2234.elec_segments

We updated data processing to ensure proper processing of assumption files (complete → approx_fun).

zgcamusa_L103.water_mapping

We corrected the data processing for completing data to fill all historical years.

**zgcamusa_L1235.elec_load_segments / zgcamusa_L1236.elec_load_segments_solver /
zgcamusa_L2235.elec_segments_FERC**

The load segment solver only attempts to assign generation by fuel to horizontal load segments in only gcamusa.LOAD_SEG_CAL_YEARS (1990, 2005, 2010, 2015) years. The logic is **very** hard coded in this file. Ideally a robust approach would be used to replace this entire methodology. Instead, in this CMP, we warn and copy forward the "solved" fuel shares to fill the historical years. However, this will throw off the load shape assumptions. Thus we subsequently need to adjust the load shape, in particular the vertical segment fractions to ensure the energy will still all balance. And finally, we make changes to ensure that the adjusted segment fraction data frame is used as input to downstream chunks.

zgcamusa_L271.nonghg_trn.R

We added safeguards against dividing by zero for PM emissions from BEV and FCEV technologies, which may not have any service output in the historical years but we do need to have emissions factors in case they will in the future.

zenergy_L154.transportation_UCD.R

We rely on vehicle intensity data from UCD / OTAQ to estimate passenger vehicle miles travelled in the historical period. Subtly this implies we want stock average intensity in the historical years and marginal intensity for future model years. The table

OTAQ_trn_data_EMF37 uses 2015 as a stock year and 2020 as a new year for vehicle intensities of vintaged technologies. If the model base year is >2015 this will result in very efficient vehicles, moreso than intended, in the final model base year. We add a stock average adjustment (assuming a 15 year turn over period) in energy_L154.transportation_UCD to the OTAQ_trn_data_EMF37 table to make this flexible for changing base years.

Update the solver configuration XML

The GCAM solver is broken out into two sections: one for calibration periods and the other for future periods. This is because we need to make sure we don't attempt to solve markets which are "fully calibrated" in the historical years and that is controlled via solution info filters. This means any time we want to run an alternative final calibration year the solver config needs to be updated accordingly.

Update Standard Policy Start Year Assumptions

Motivated by the new model base year we push back by 5 years to 2030 the start of all of our standard validation policy assumptions. With the exception of SSP 2 and 5 which continue to start global coordination in 2040.

Drop "climateOutputInterval" from configurations / reporting code

The configuration "int" `climateOutputInterval` would control the frequency that annual results such as climate model outputs or land use change emissions would be reported to output including the XMLDB. Given a simple constant interval would not be able to sufficiently report when we have non-constant model time steps such as for a 2021 model year. Instead, we just always write annual results from the C++. However, because this would be a lot of data for LUC emissions we update the `default_filter.xml` to filter out results in non-model years. Given, the logic has been moved out of the C++ and into a user modifiable filter script this provides us much more flexibility to cover user needs.

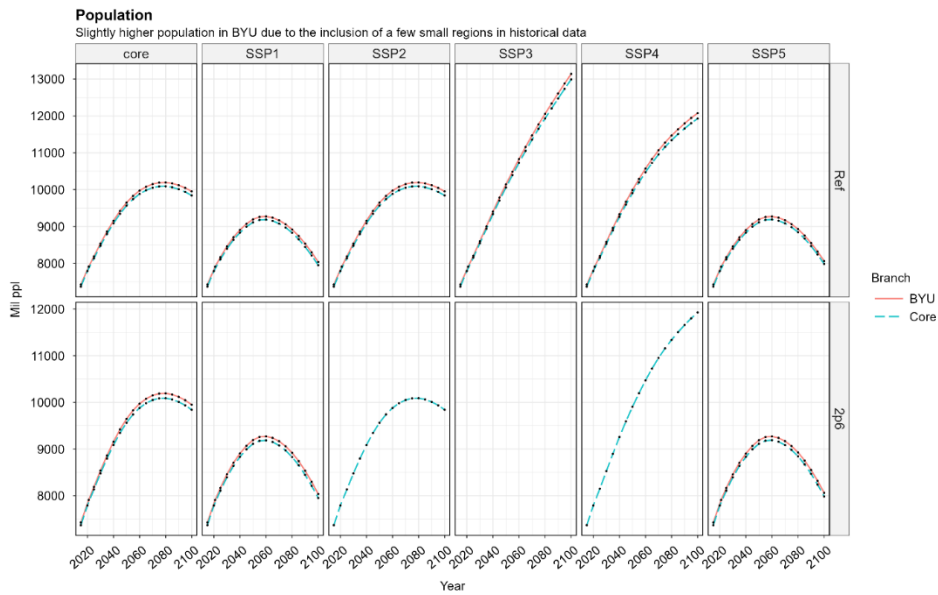
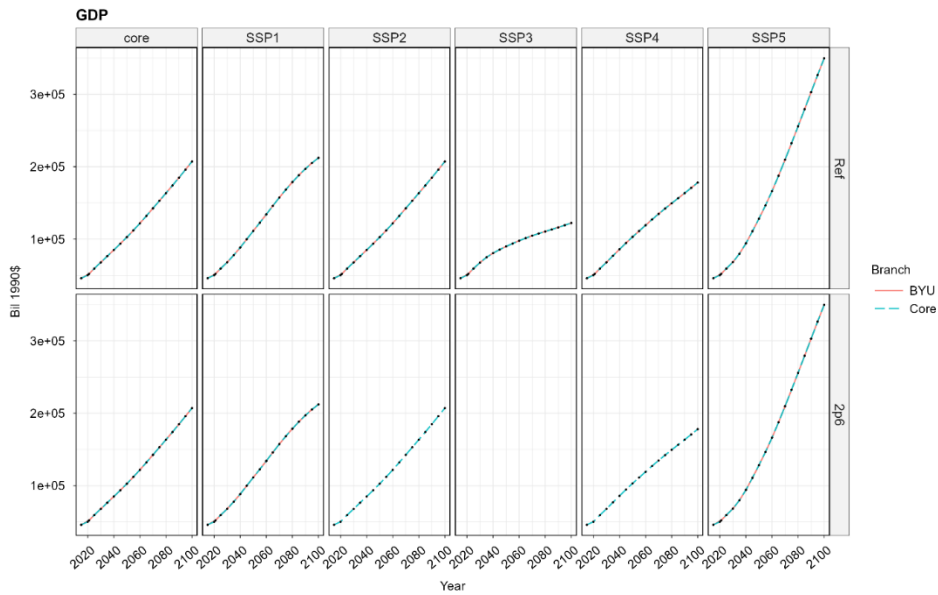
Fix Other Industry Income Elasticity

Fix [bug reported by the community on GitHub](#) where other industry income elasticities were much higher than intended. This issue came about because the income elasticities were not a simple assumption but rather a look up function relating total energy -> income elasticity (more rapid growth at lower energy). Circa GCAM 6/7 we broke out many "detailed industries" and moved out > 50% of the energy from the "other industry" sector. But we forgot to update the other industry income elasticity lookup assumption CSV. Ultimately this resulted in more than 100 EJ of primary energy than otherwise intended and is discussed further in the validation.

Validation

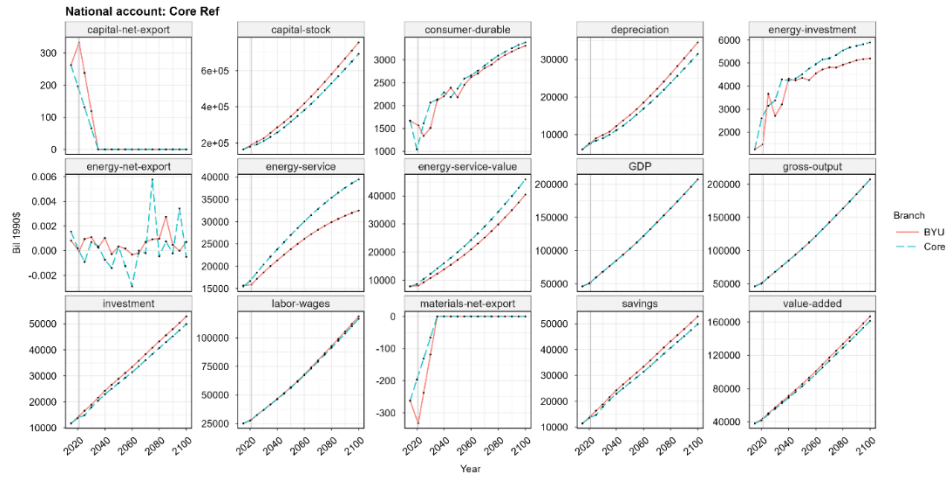
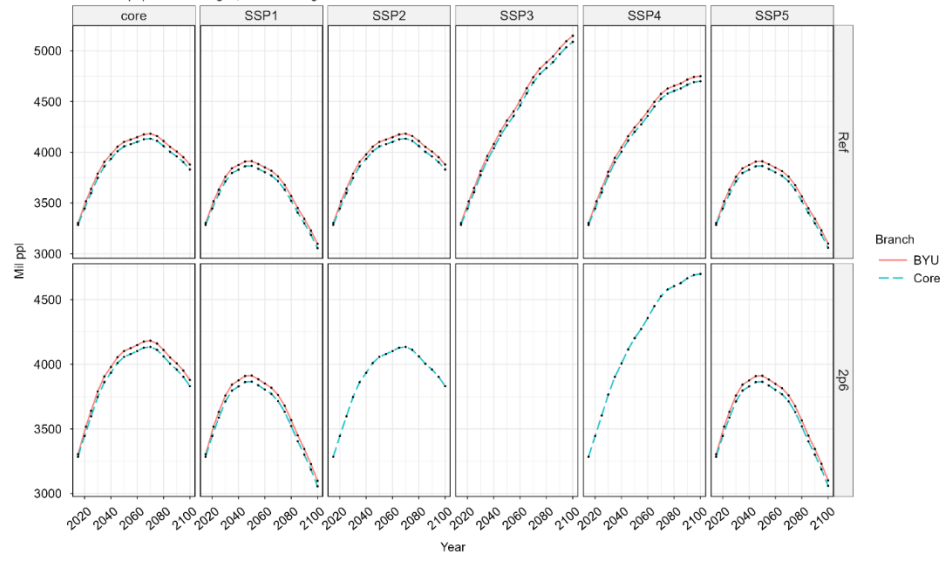
Naturally updating the final calibration year has wide ranging effects on model results. As such we provide an extensive, yet high level, overview of the change in results. It should be noted that also playing a factor in these changes is the bug fix for other industry which itself ultimately results in a large reduction in primary energy, on the magnitude of ~ 100 EJ by 2100.

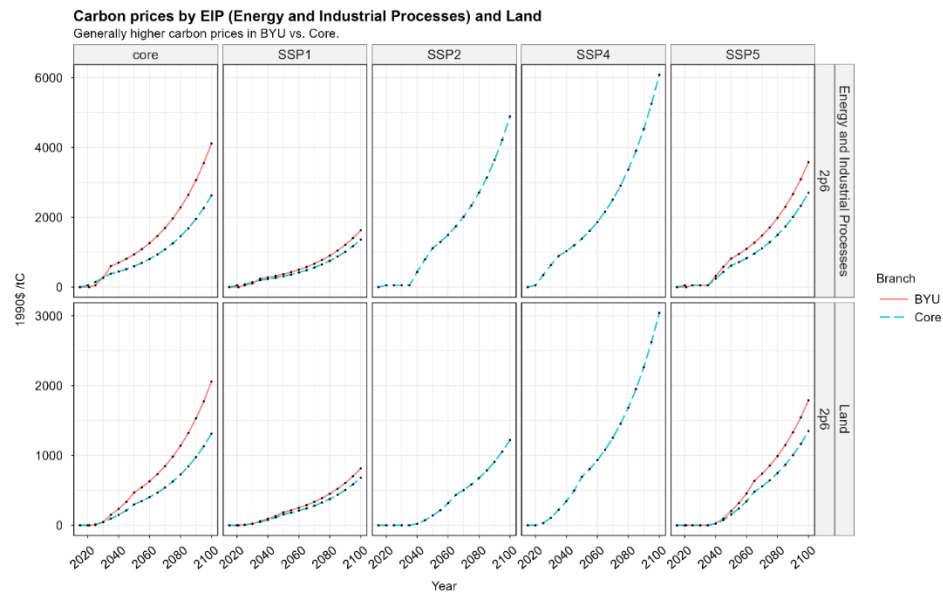
Socioeconomics



Labor force

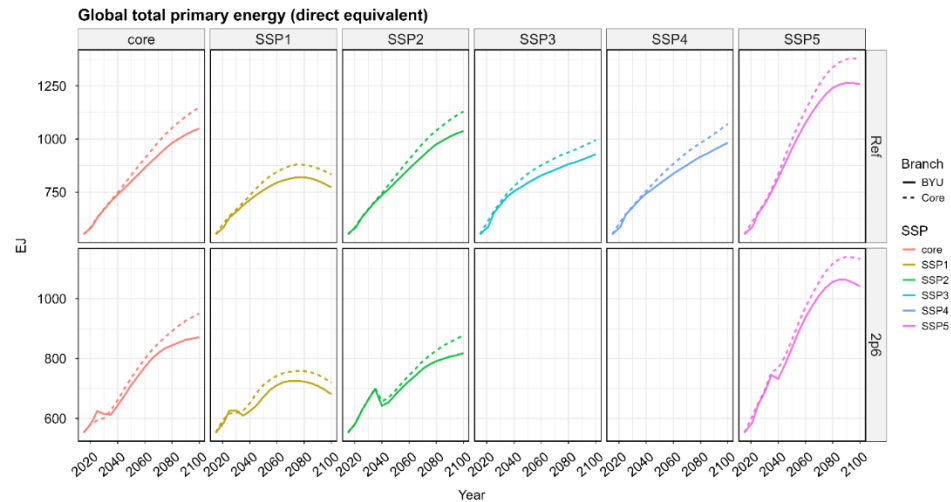
Similar to population changes, there was higher labor force



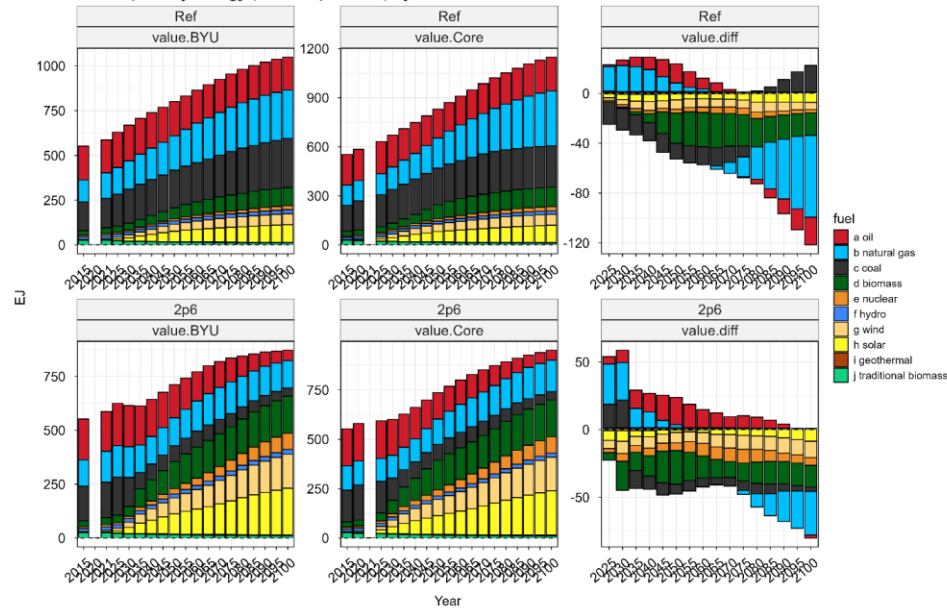


Energy

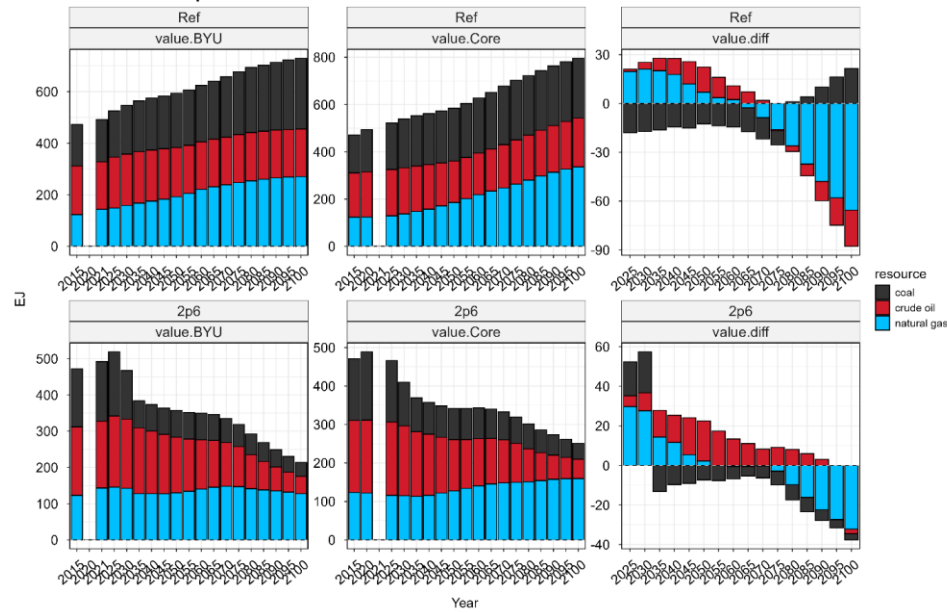
Primary Energy and Energy Trade



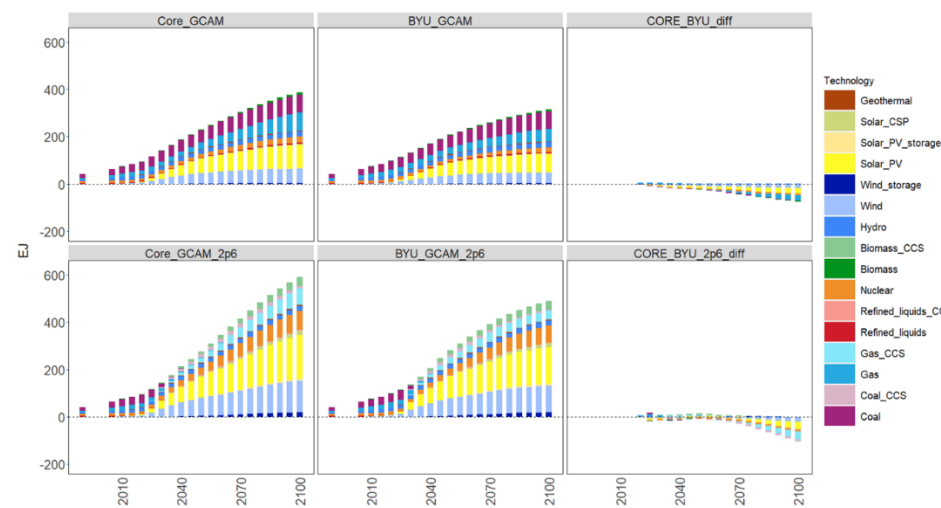
Global primary energy (direct equivalent) by fuel



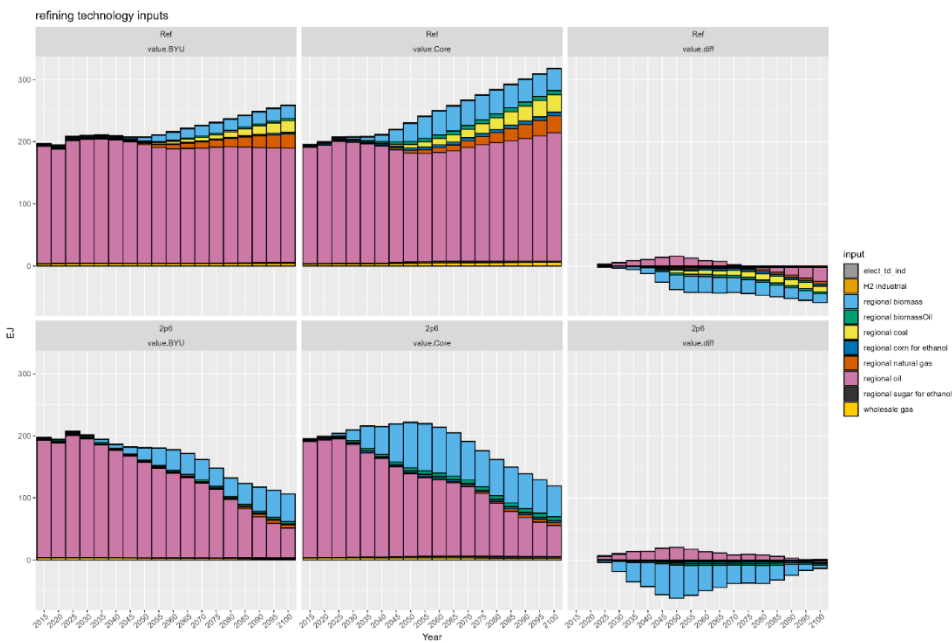
Global resource production



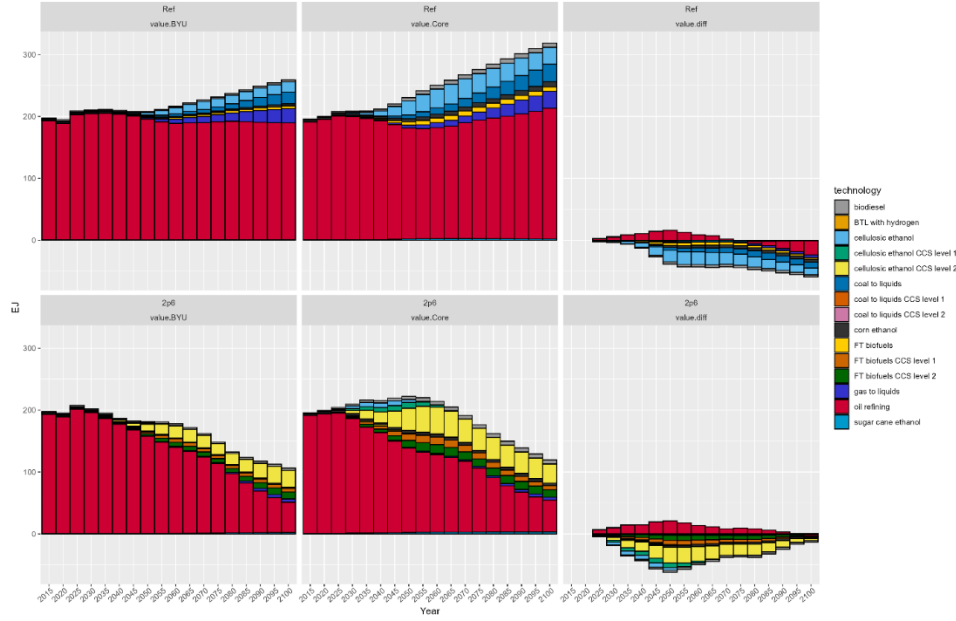
Electricity



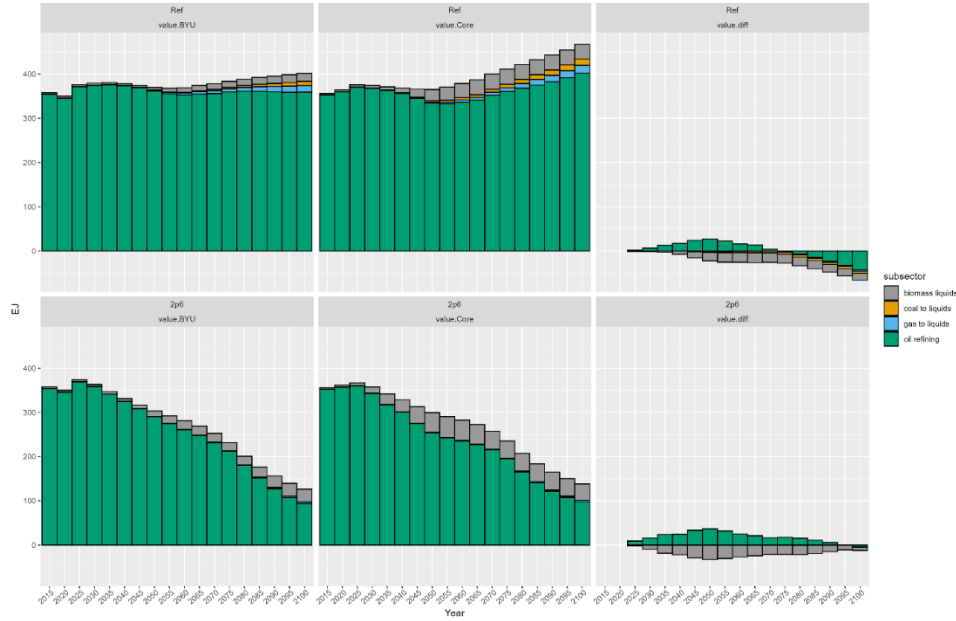
Refining



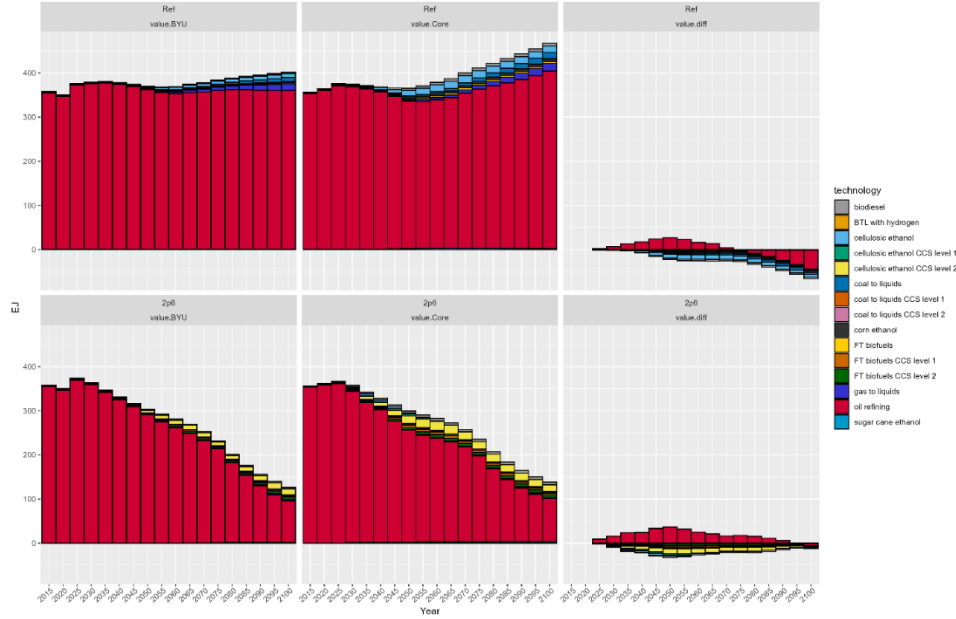
refining inputs by technology



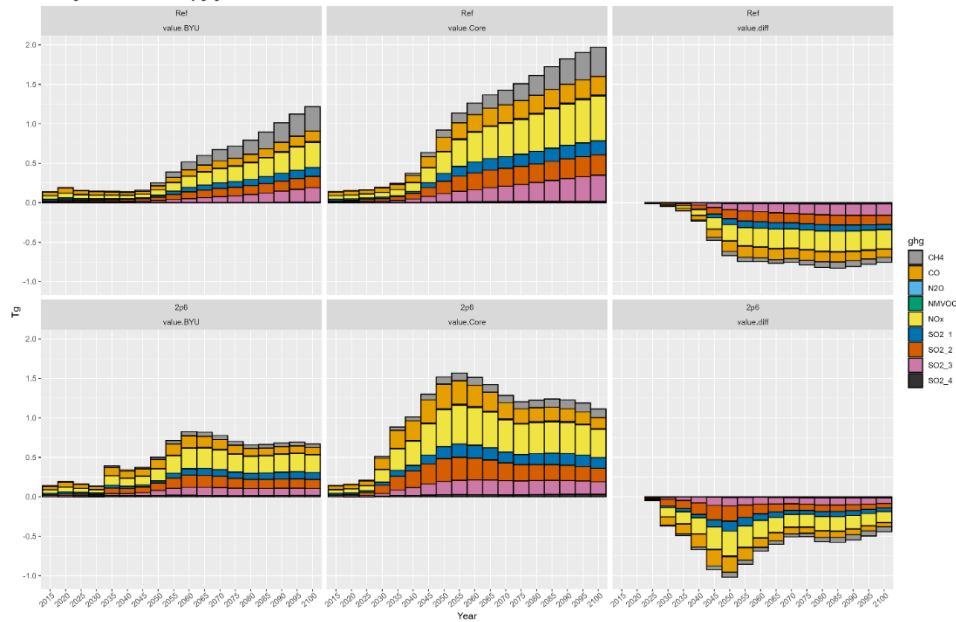
refining outputs by subsector



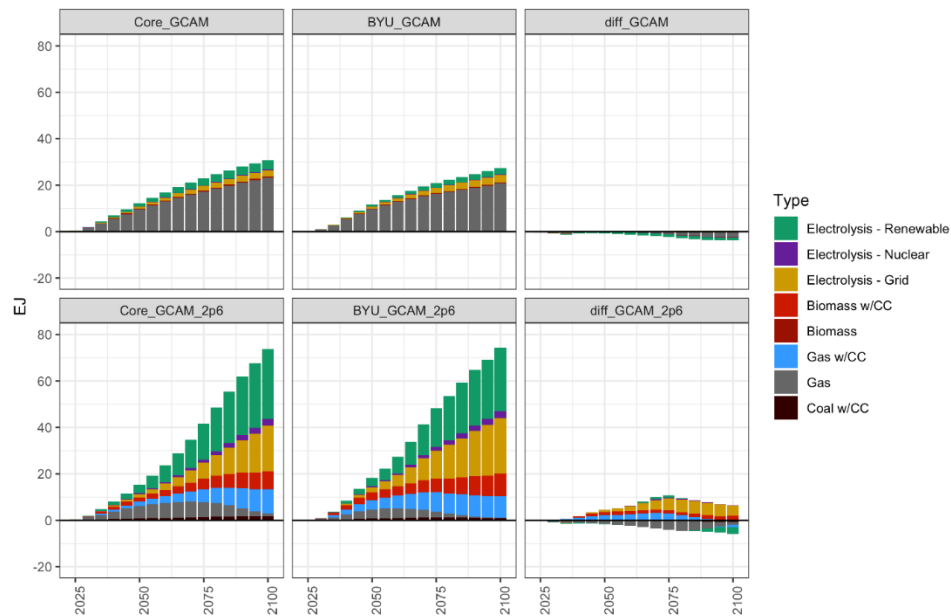
refining outputs by technology



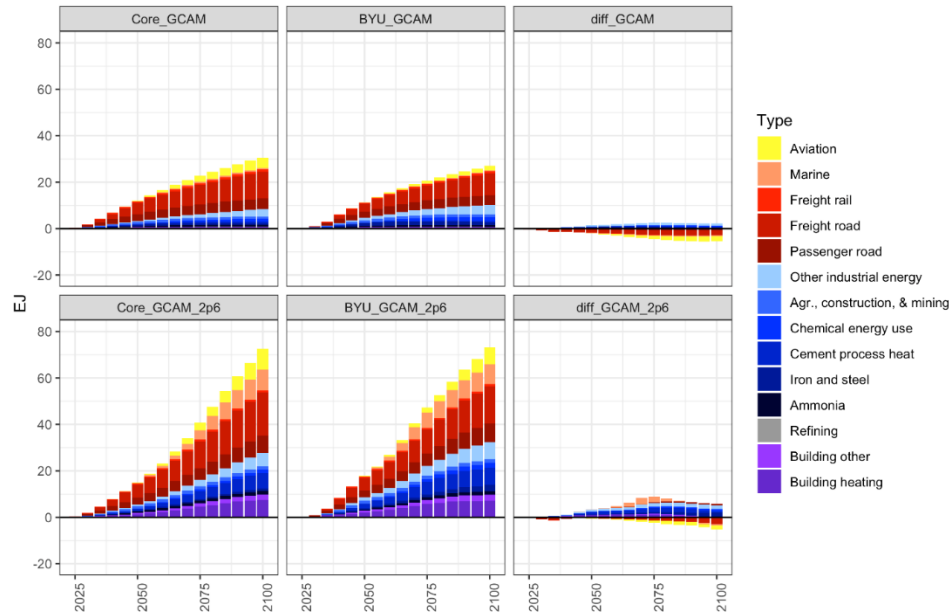
refining nonCO2 emissions by ghg



Hydrogen

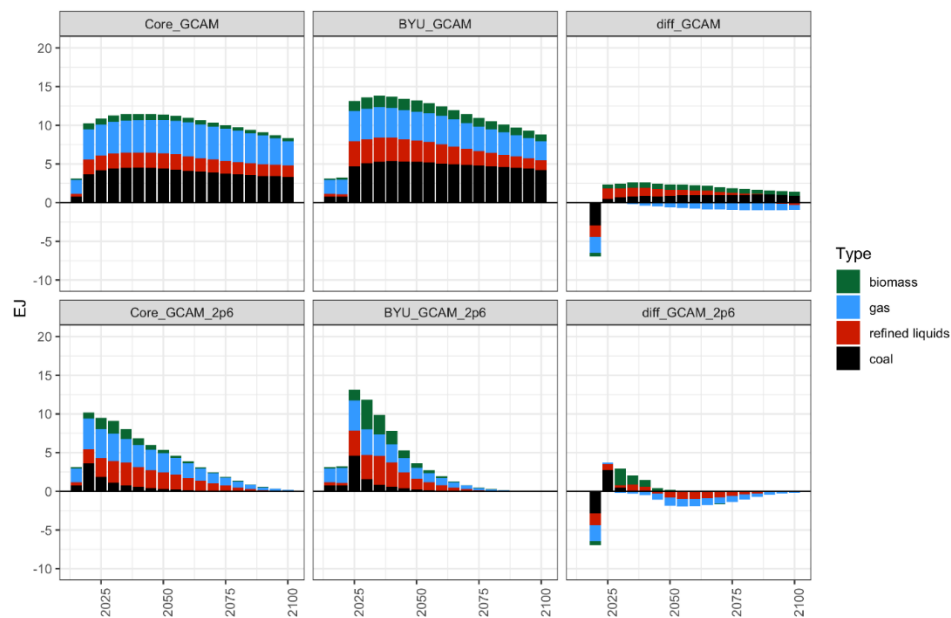


Global hydrogen production by type. Differences are shown for the base year update proposal relative to the core version of GCAM.

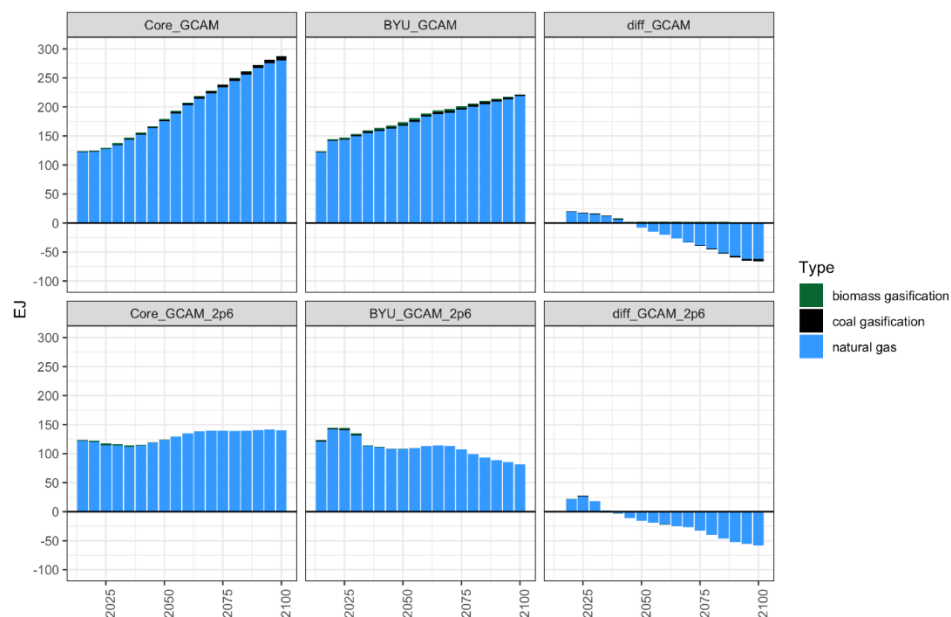


Global hydrogen consumption by sector. Differences are shown for the base year update proposal relative to the core version of GCAM.

Other Energy Transformation (District Heat, Gas Processing)



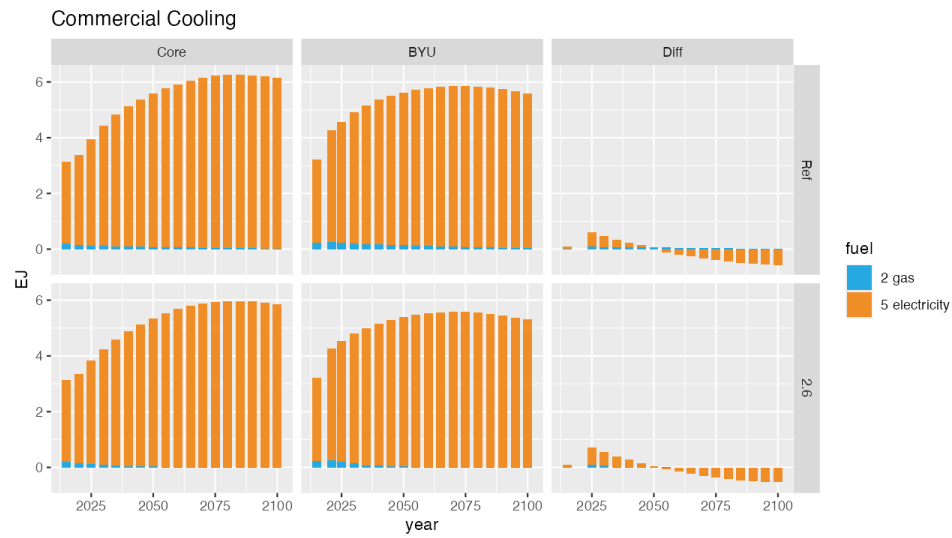
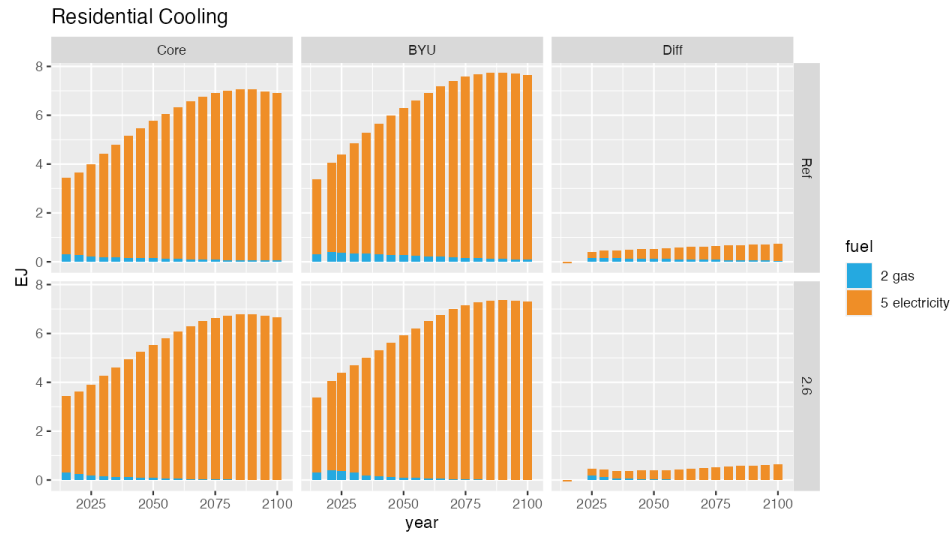
Global district heat production by fuel. Differences are shown for the base year update proposal relative to the core version of GCAM. Data for 2021 in the base year update is labelled as "2020" for the figure.

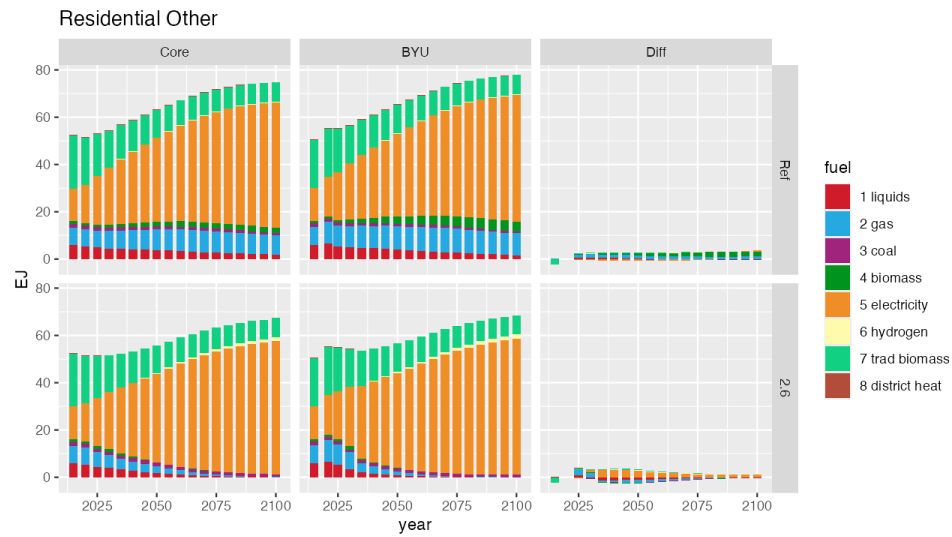


Global gas processing by type. Differences are shown for the base year update proposal relative to the core version of GCAM. Data for 2021 in the base year update is labelled as "2020" for the figure.

Buildings

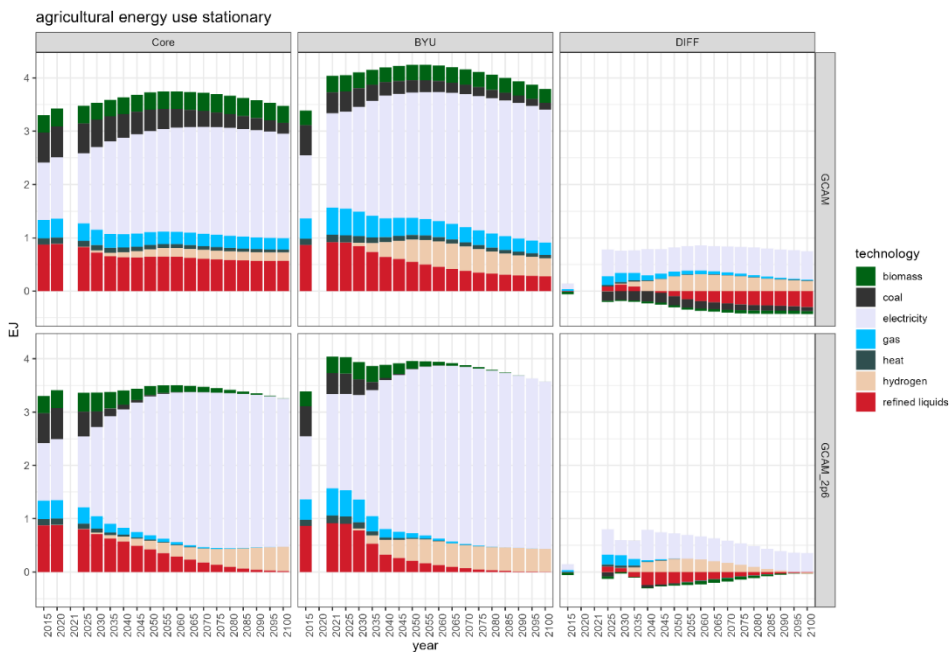
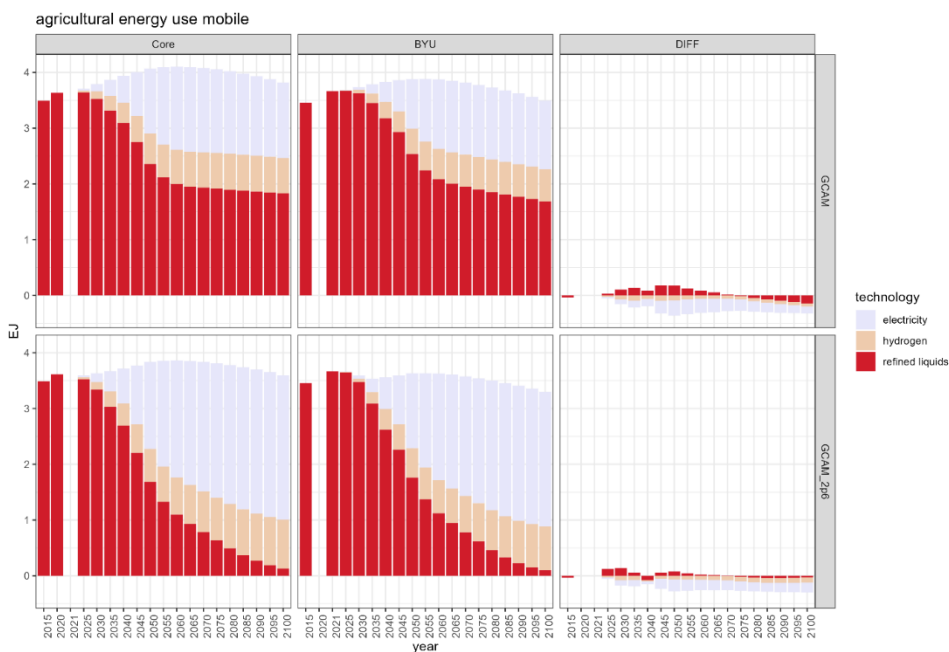


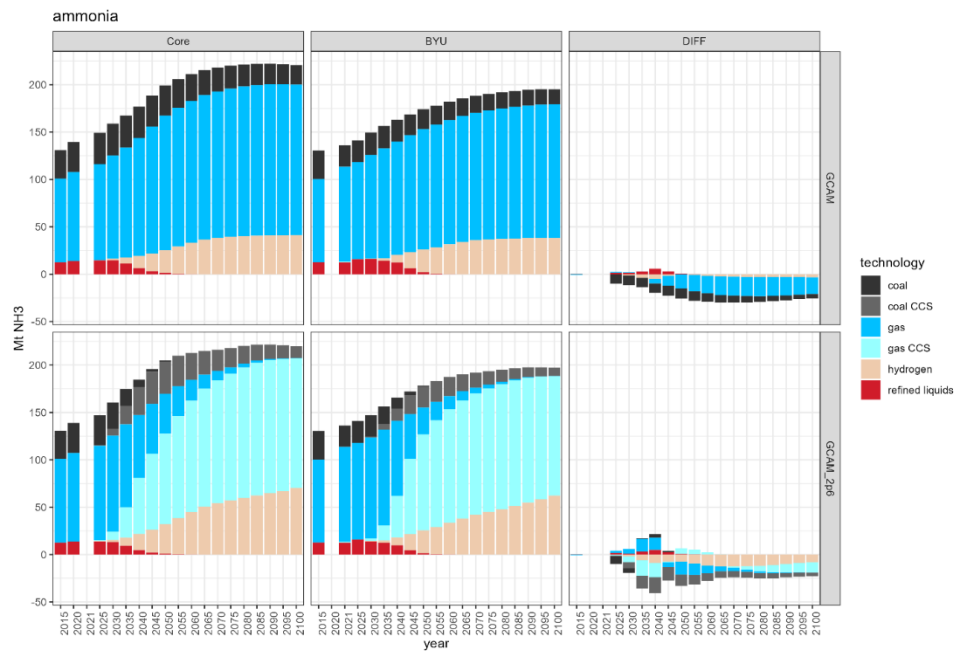
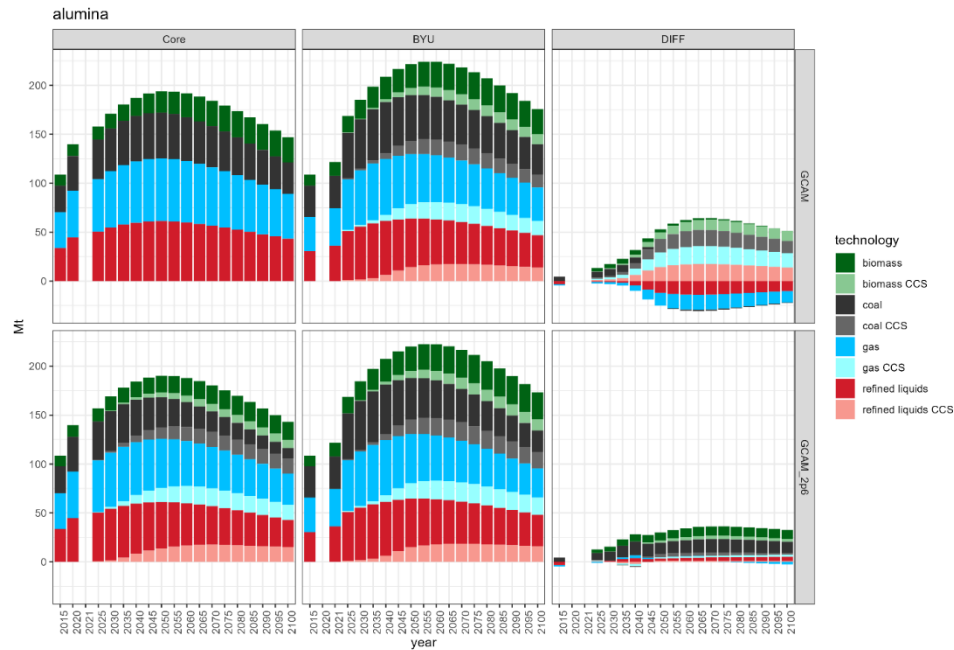


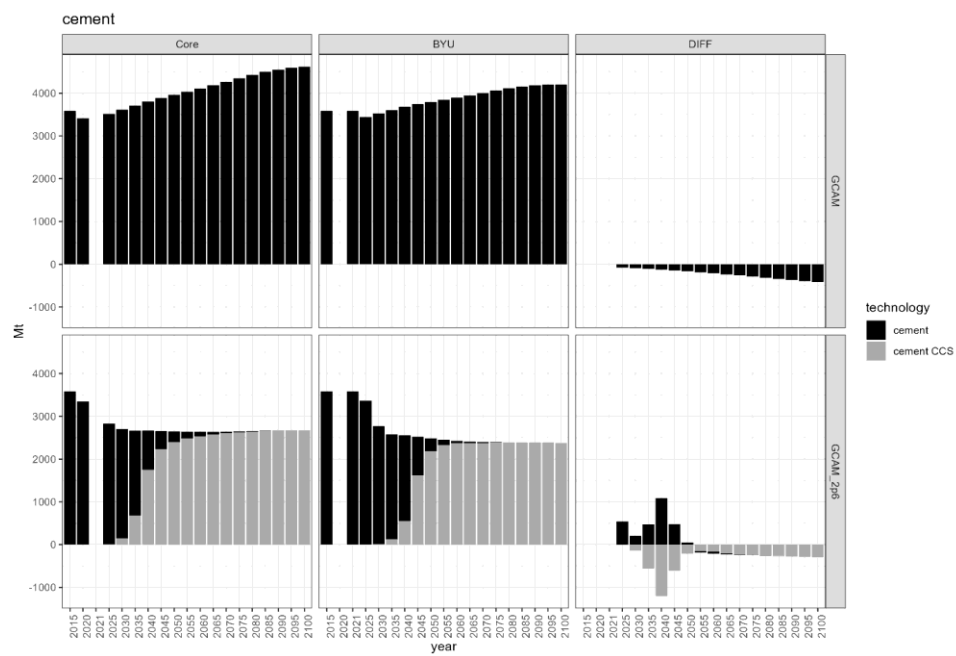
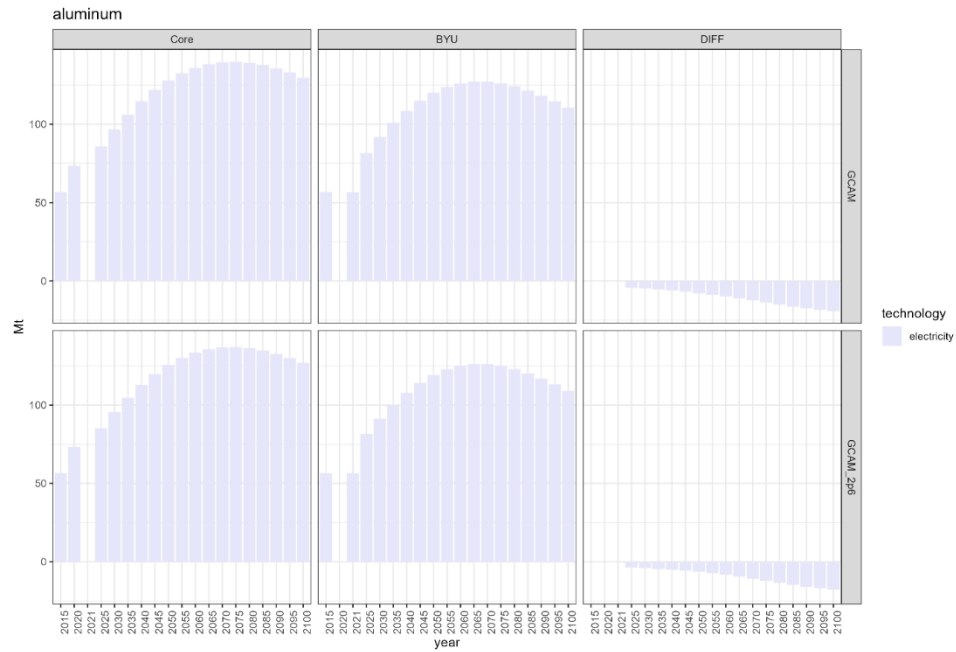


Industry

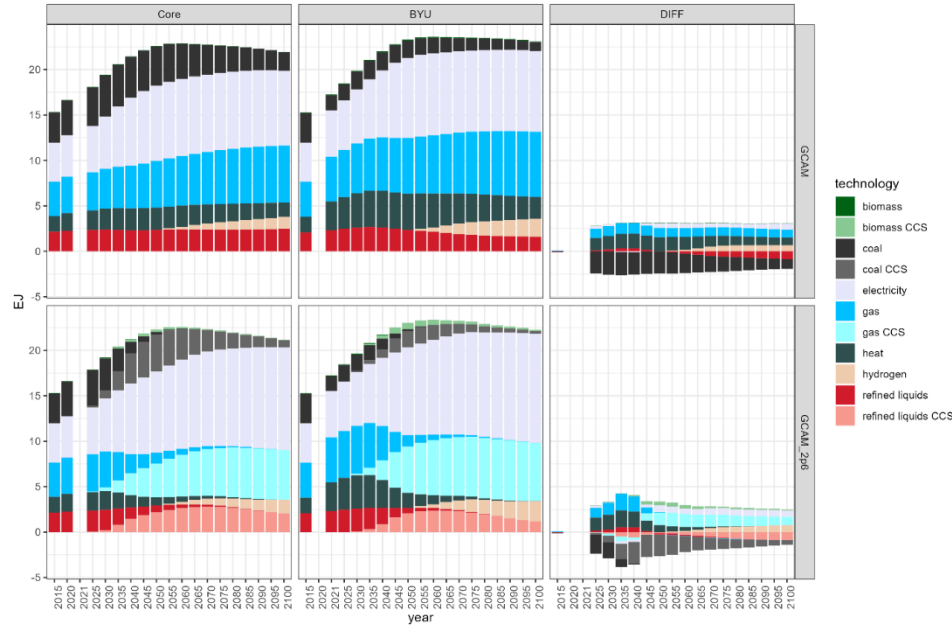
Industry Outputs by Sector and Technology



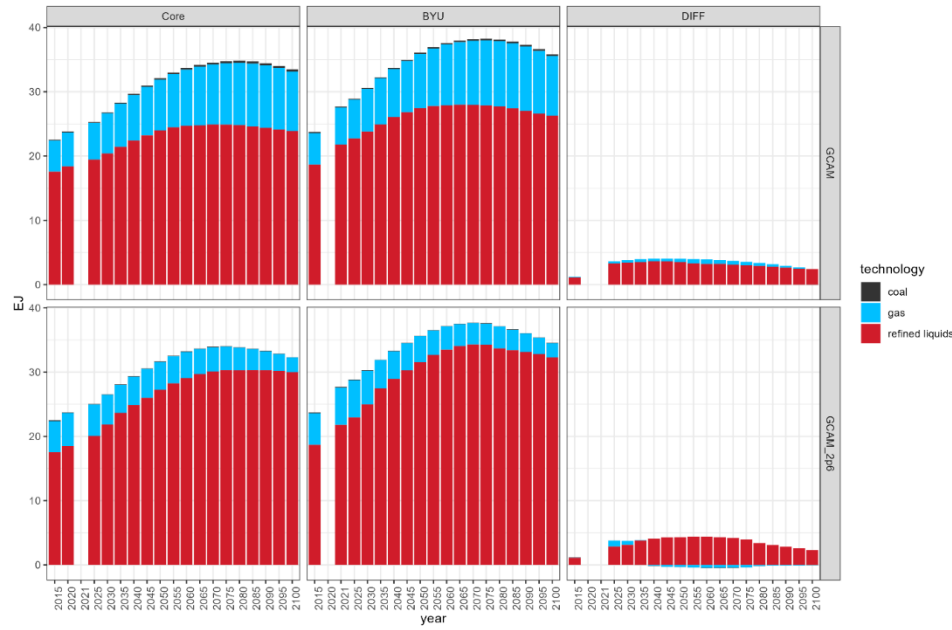




chemical energy use



chemical feedstocks



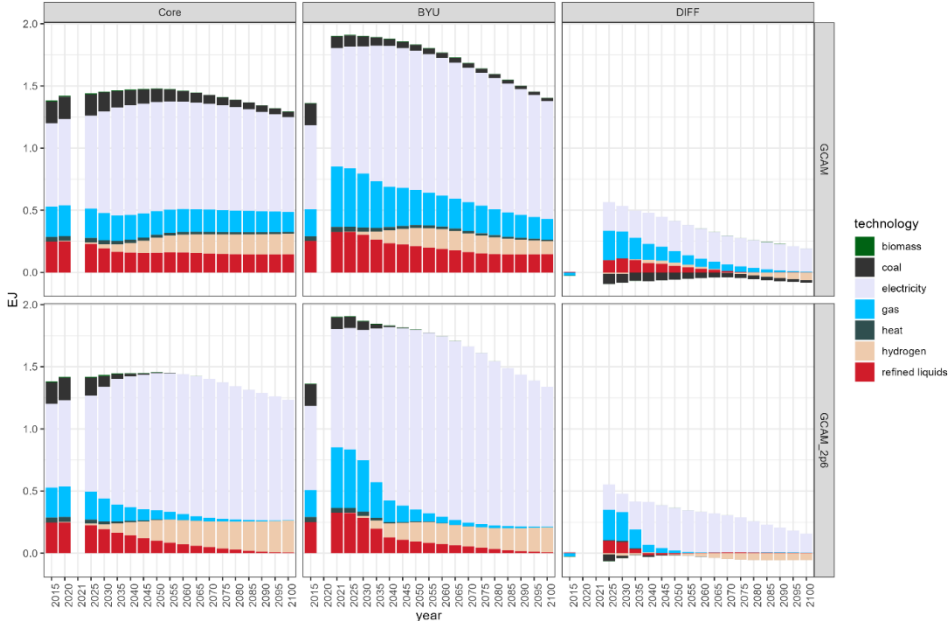
construction energy use mobile



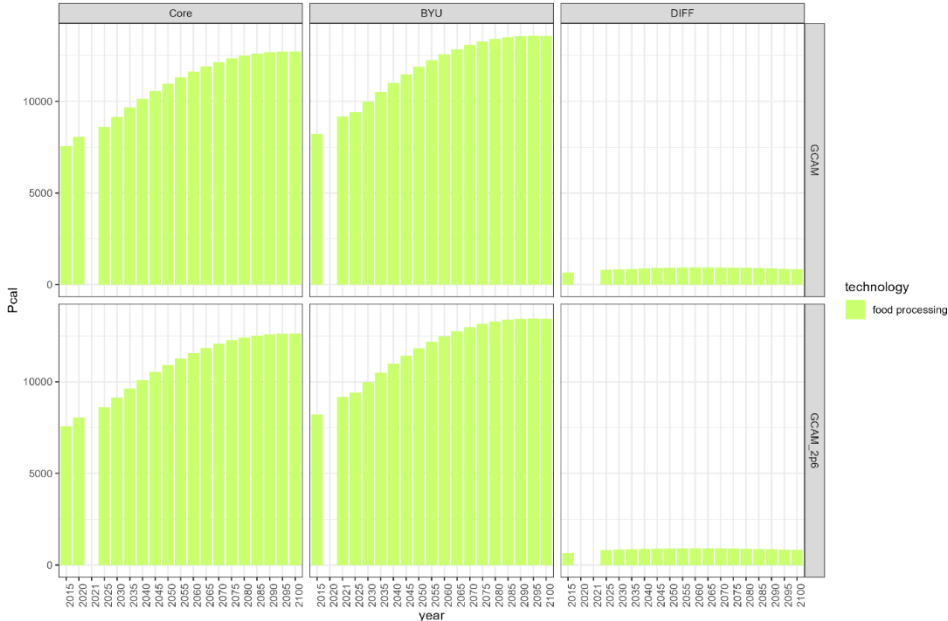
construction feedstocks



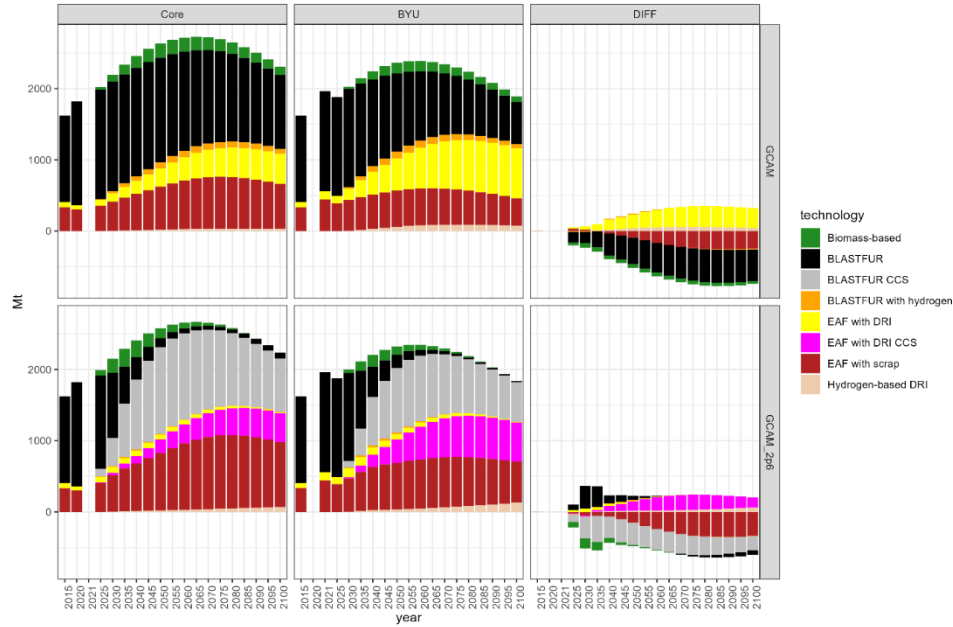
construction energy use stationary



food processing



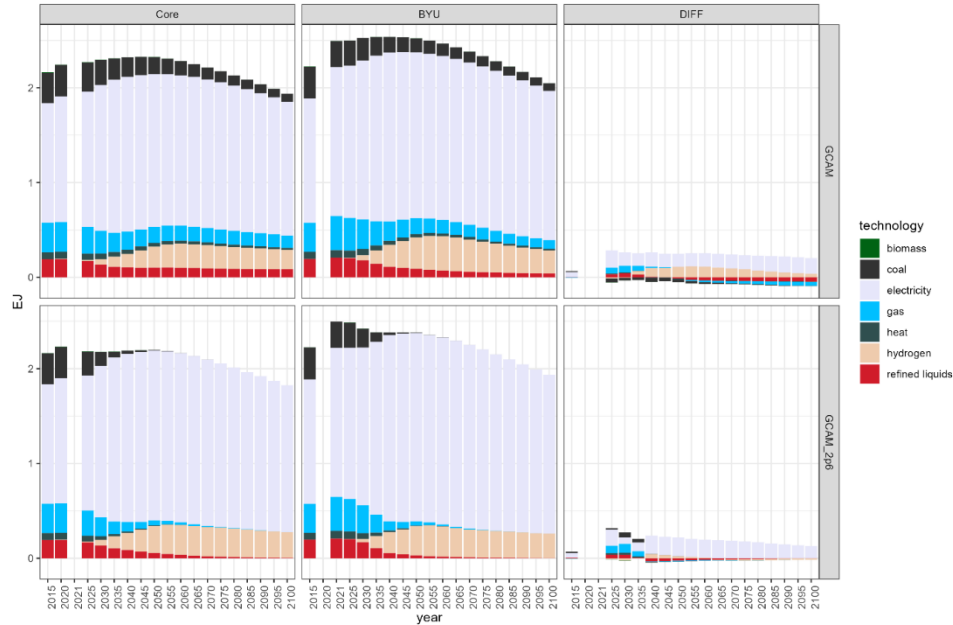
iron and steel



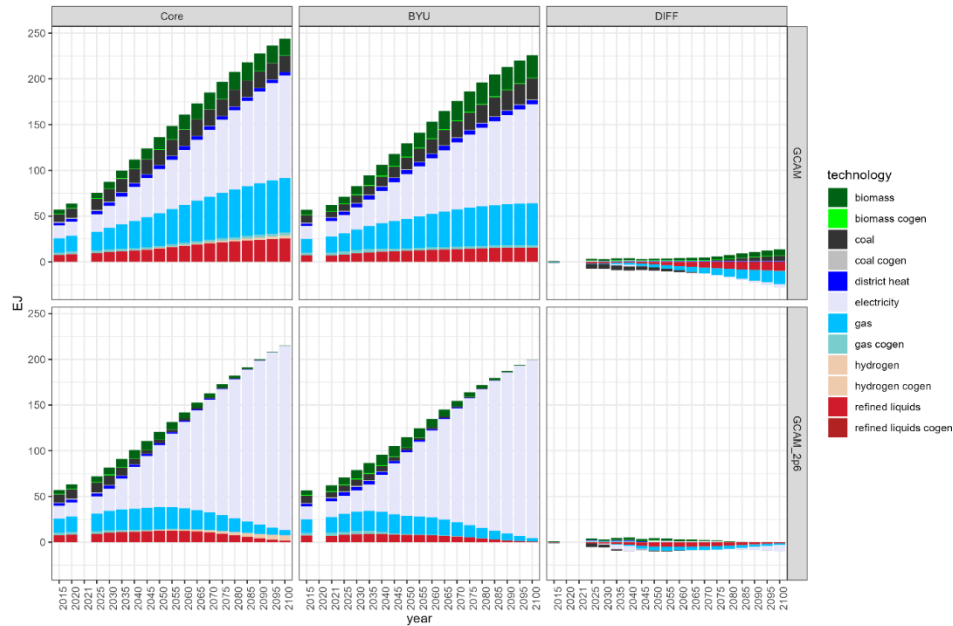
mining energy use mobile



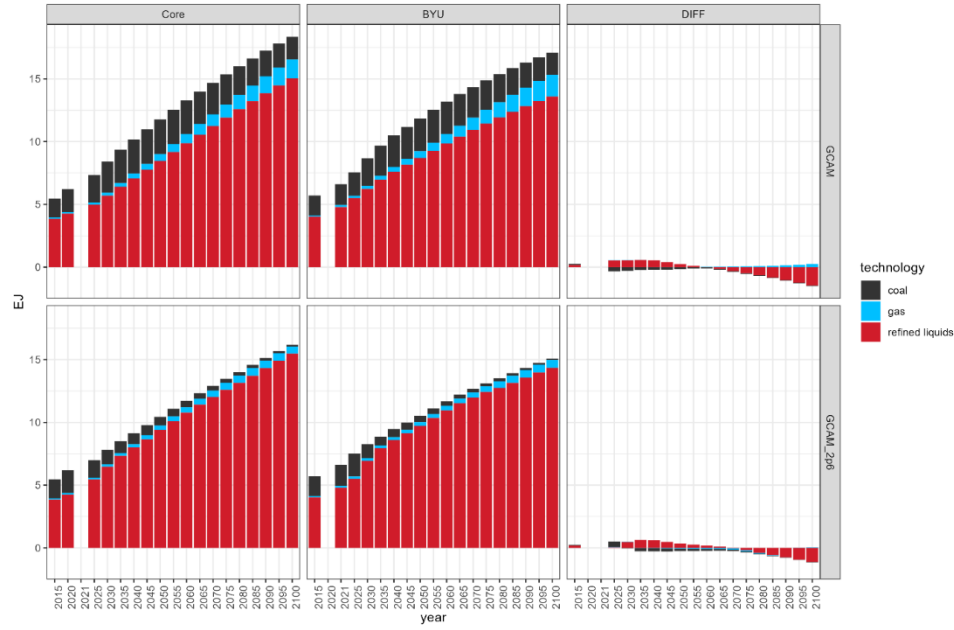
mining energy use stationary



other industrial energy use



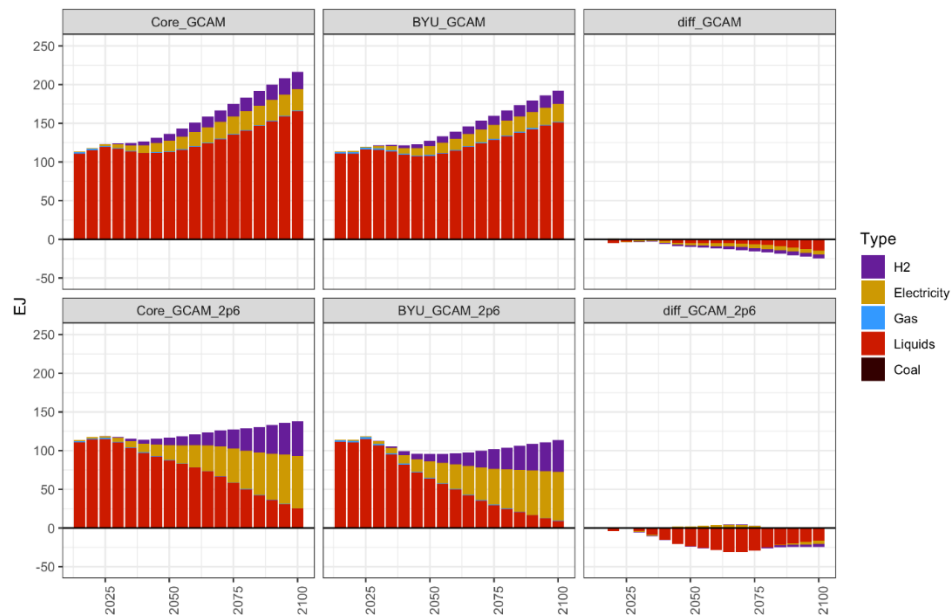
other industrial feedstocks



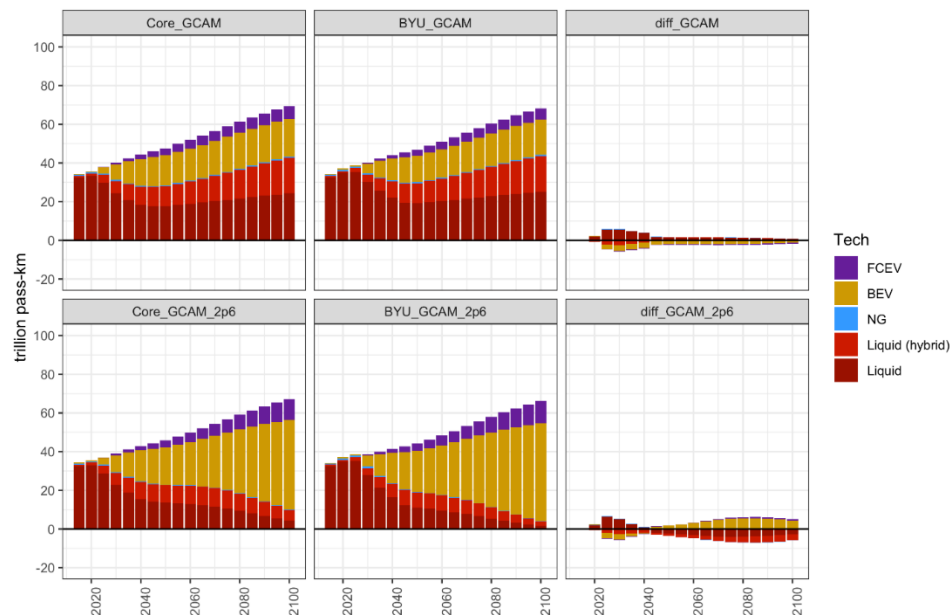
paper



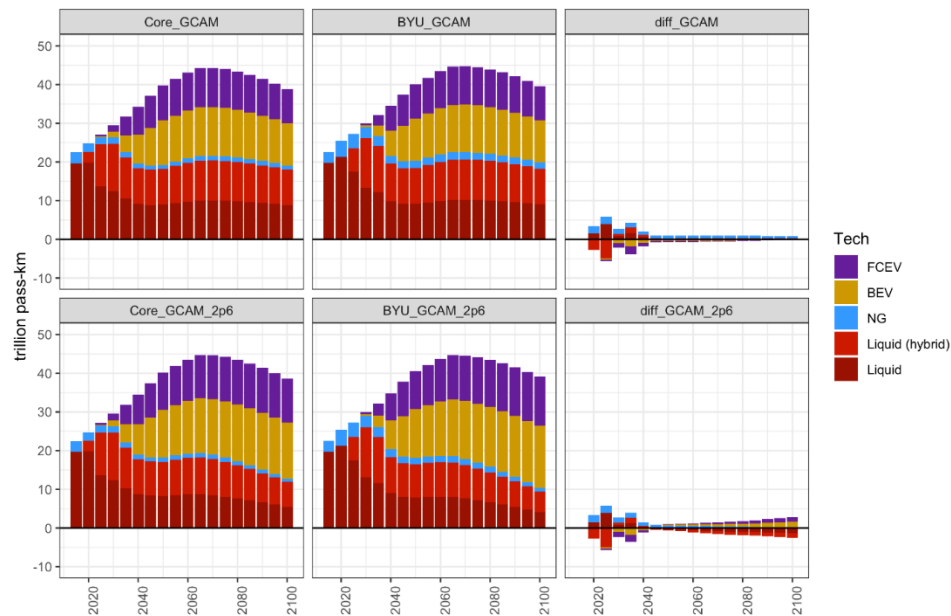
Transport



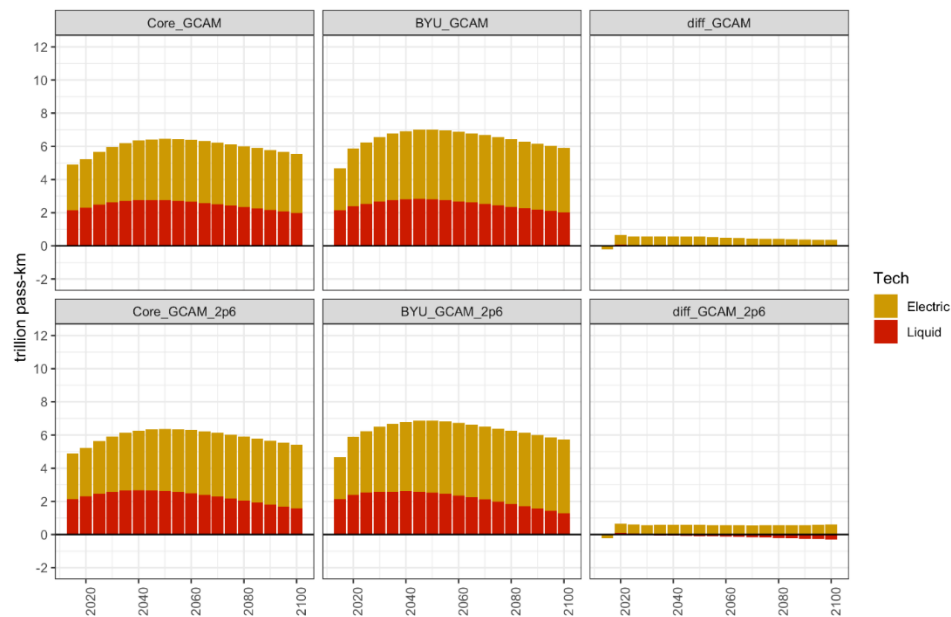
Global transportation final energy by fuel. Differences are shown for the base year update proposal relative to the core version of GCAM. Data for 2021 in the base year update is labelled as "2020" for the figure.



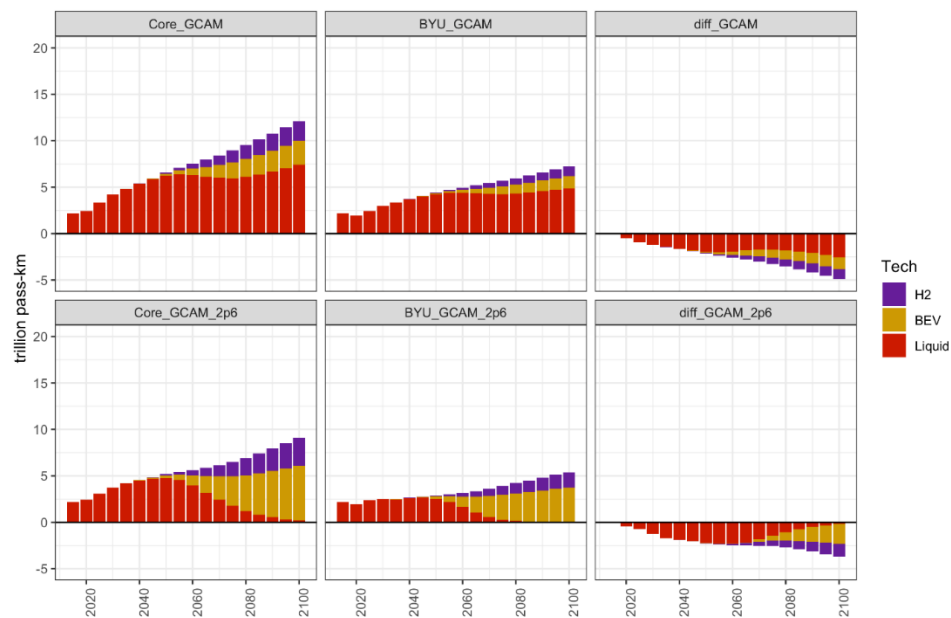
Global passenger road service output by type. Differences are shown for the base year update proposal relative to the core version of GCAM. Data for 2021 in the base year update is labelled as "2020" for the figure. This figure includes vehicles with 2 or 3 wheels as well as passenger cars.



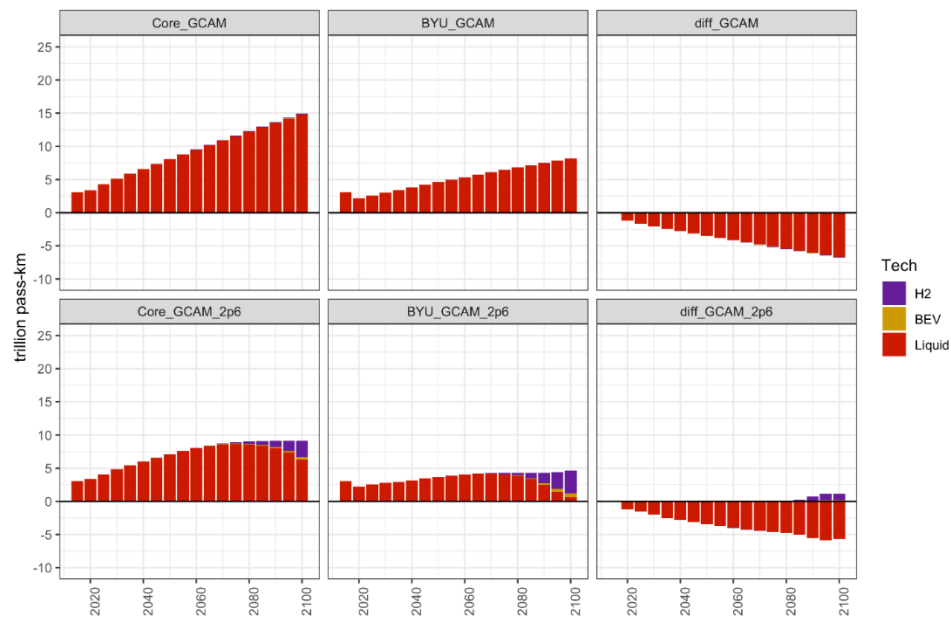
Global bus service output by type. Differences are shown for the base year update proposal relative to the core version of GCAM. Data for 2021 in the base year update is labelled as "2020" for the figure.



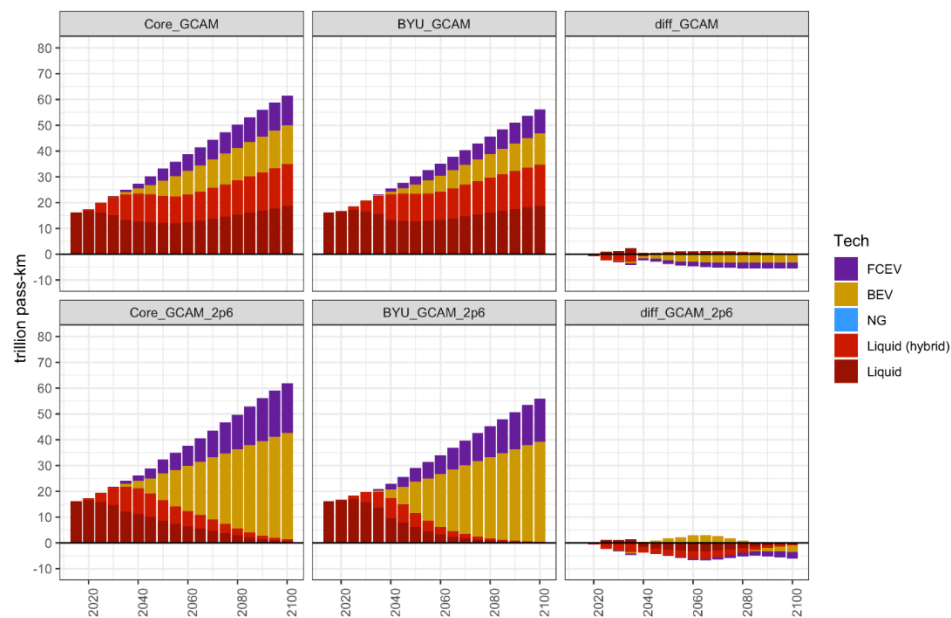
Global passenger rail service output by type. Differences are shown for the base year update proposal relative to the core version of GCAM. Data for 2021 in the base year update is labelled as "2020" for the figure.



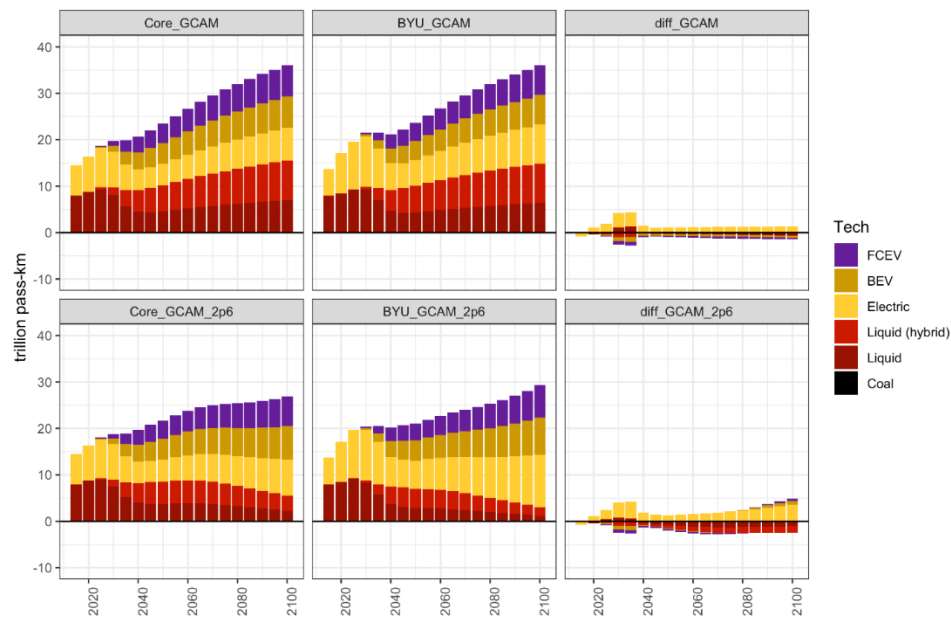
Global domestic aviation service output by type. Differences are shown for the base year update proposal relative to the core version of GCAM. Data for 2021 in the base year update is labelled as "2020" for the figure.



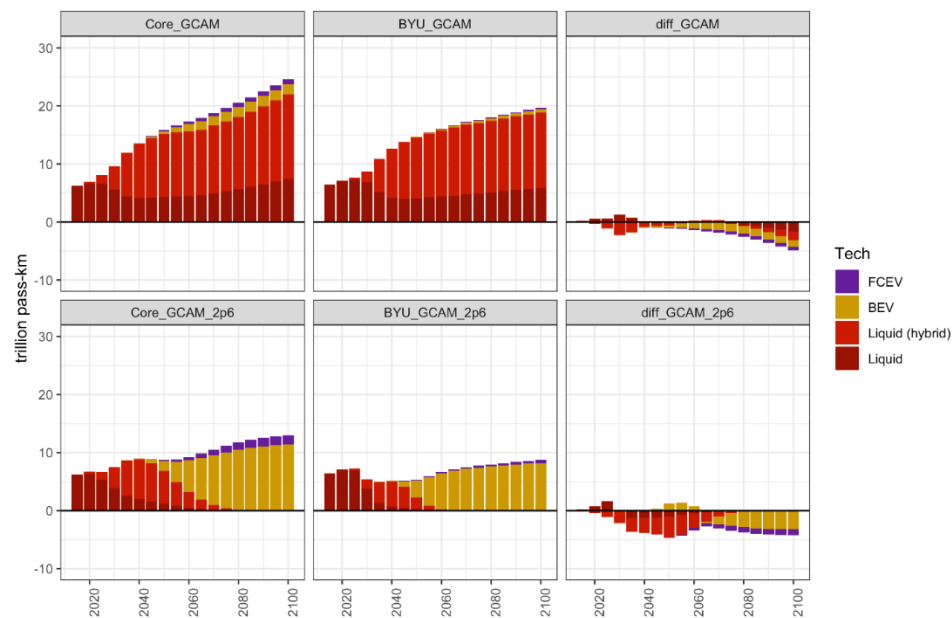
Global international aviation service output by type. Differences are shown for the base year update proposal relative to the core version of GCAM. Data for 2021 in the base year update is labelled as "2020" for the figure.



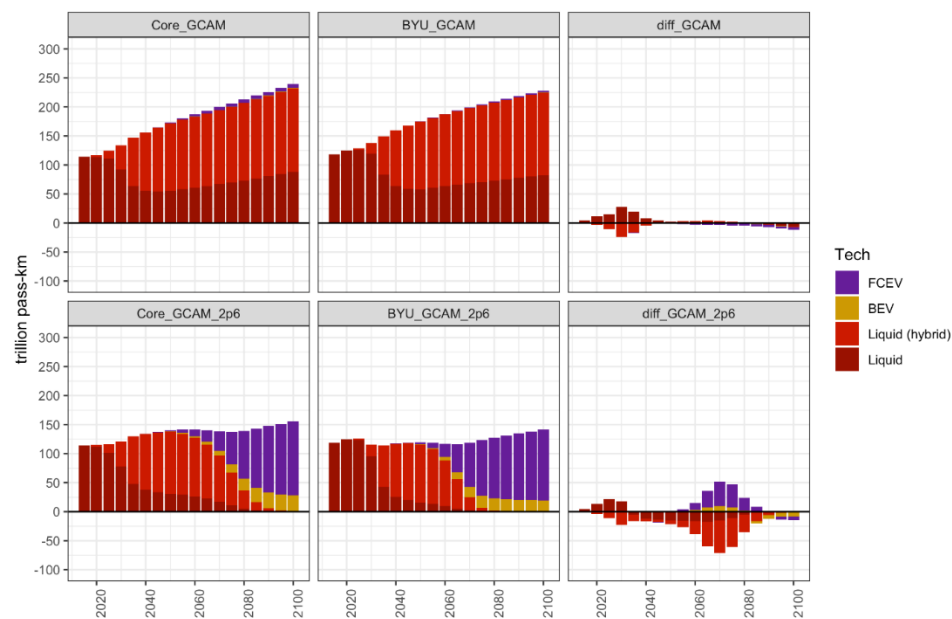
Global freight road service output by type. Differences are shown for the base year update proposal relative to the core version of GCAM. Data for 2021 in the base year update is labelled as "2020" for the figure.



Global freight rail service output by type. Differences are shown for the base year update proposal relative to the core version of GCAM. Data for 2021 in the base year update is labelled as "2020" for the figure.



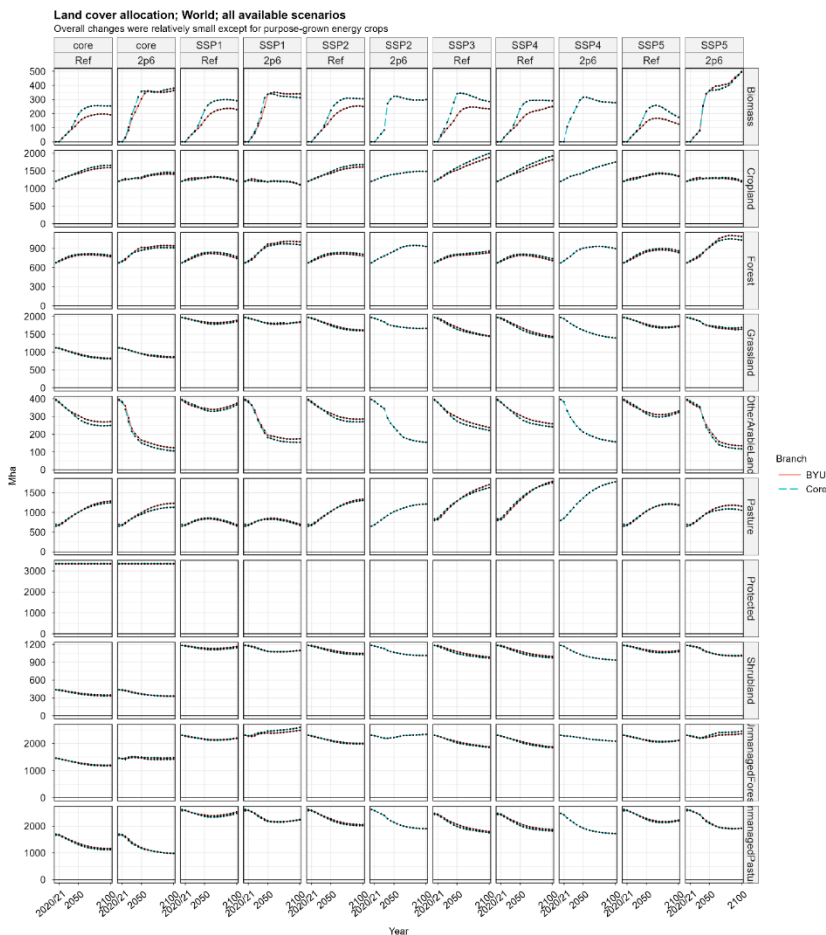
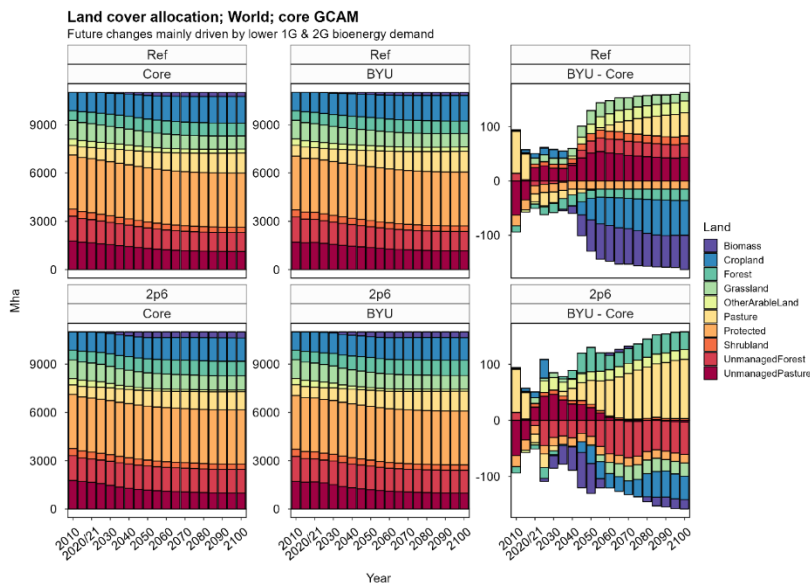
Global domestic shipping service output by type. Differences are shown for the base year update proposal relative to the core version of GCAM. Data for 2021 in the base year update is labelled as "2020" for the figure.



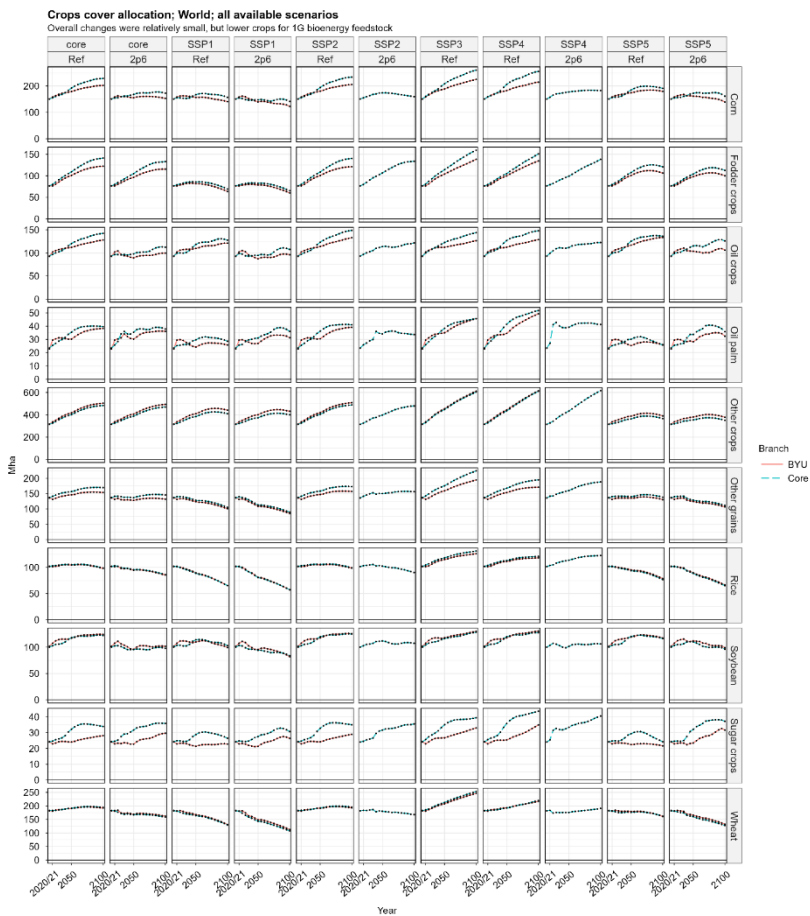
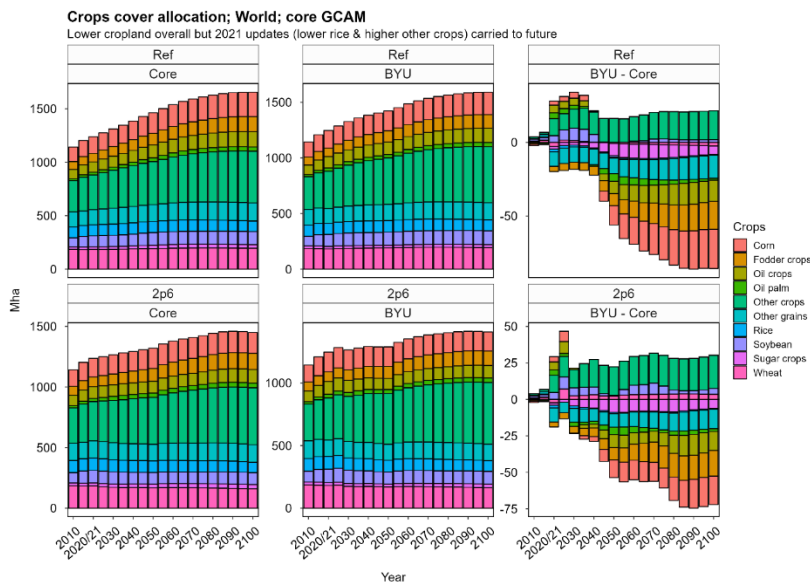
Global international shipping service output by type. Differences are shown for the base year update proposal relative to the core version of GCAM. Data for 2021 in the base year update is labelled as "2020" for the figure.

Agriculture and Land Use

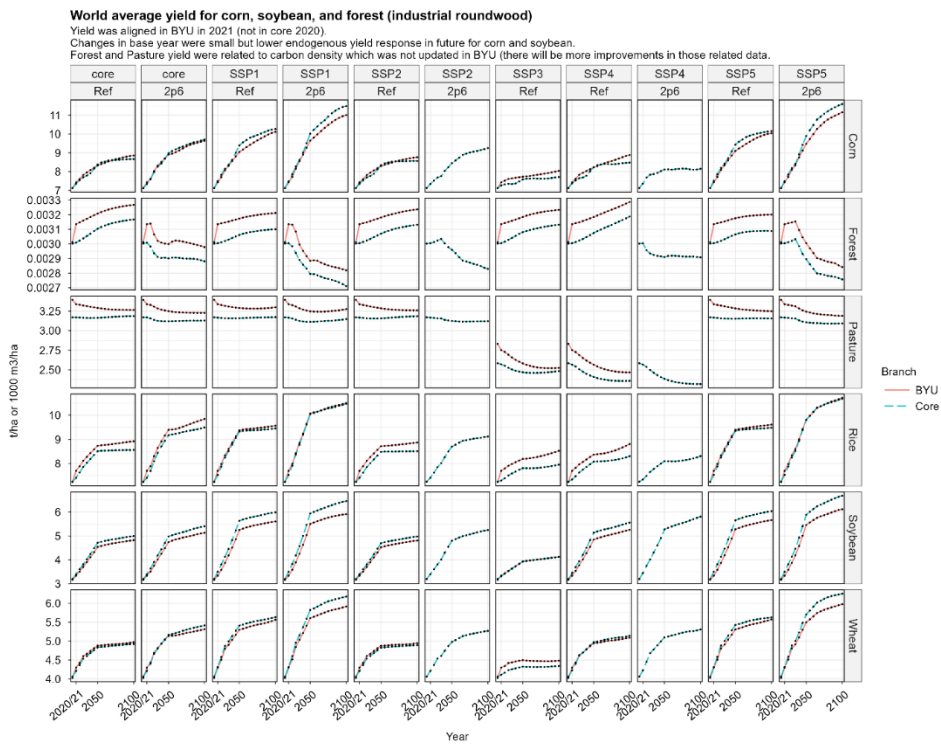
Land cover



Cropland by crops

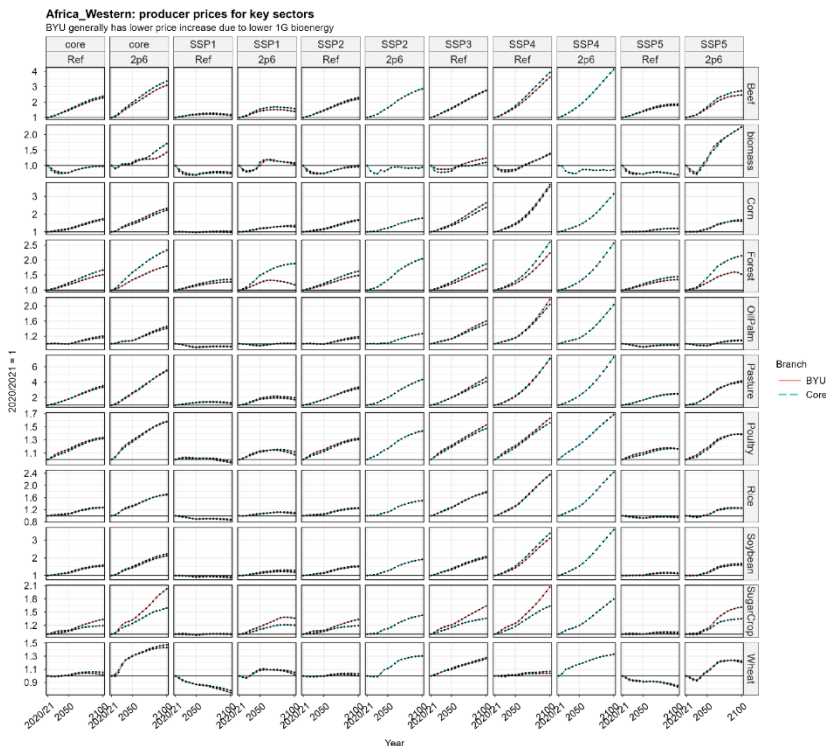
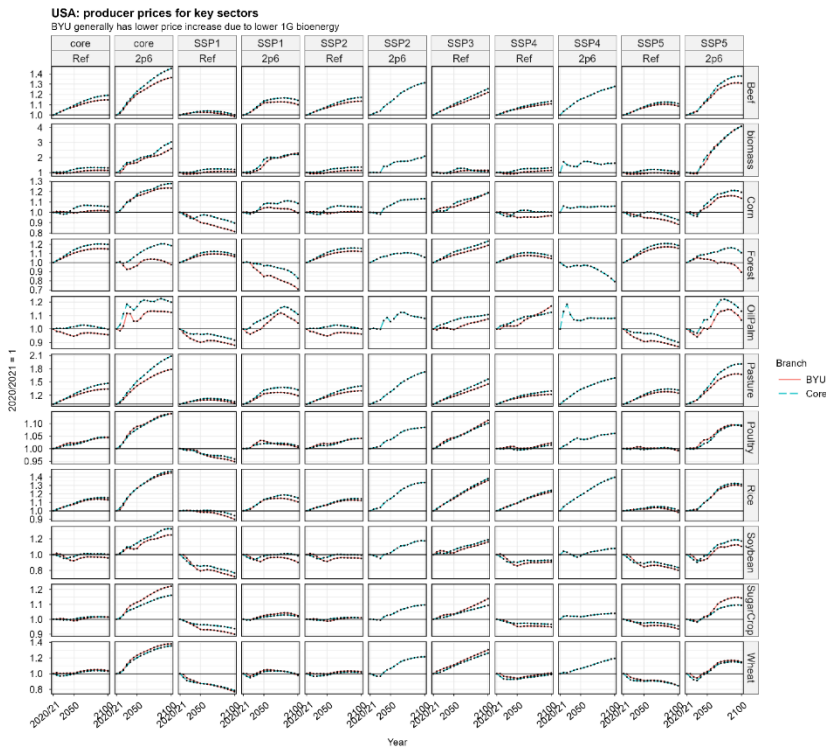


Yield



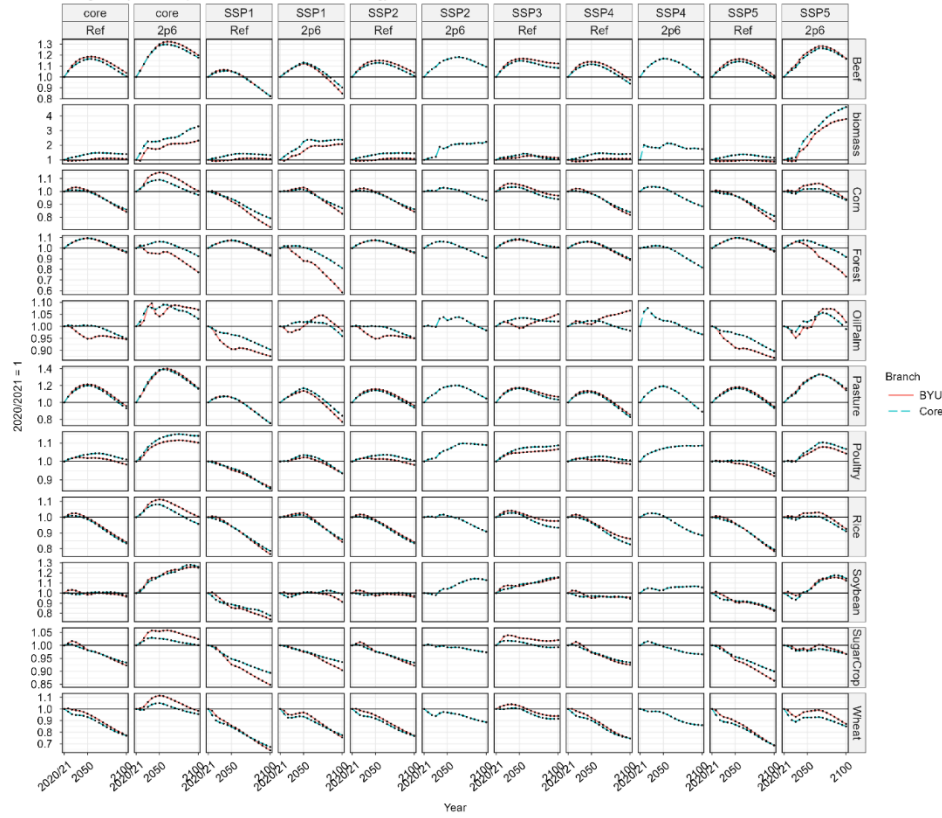
Producer prices

The trend is generally consistent after BYU. However, the 2021 prices (in \$/kg) are generally higher than those projected in Core due to the unrecognized disruptions (COVID& wars).



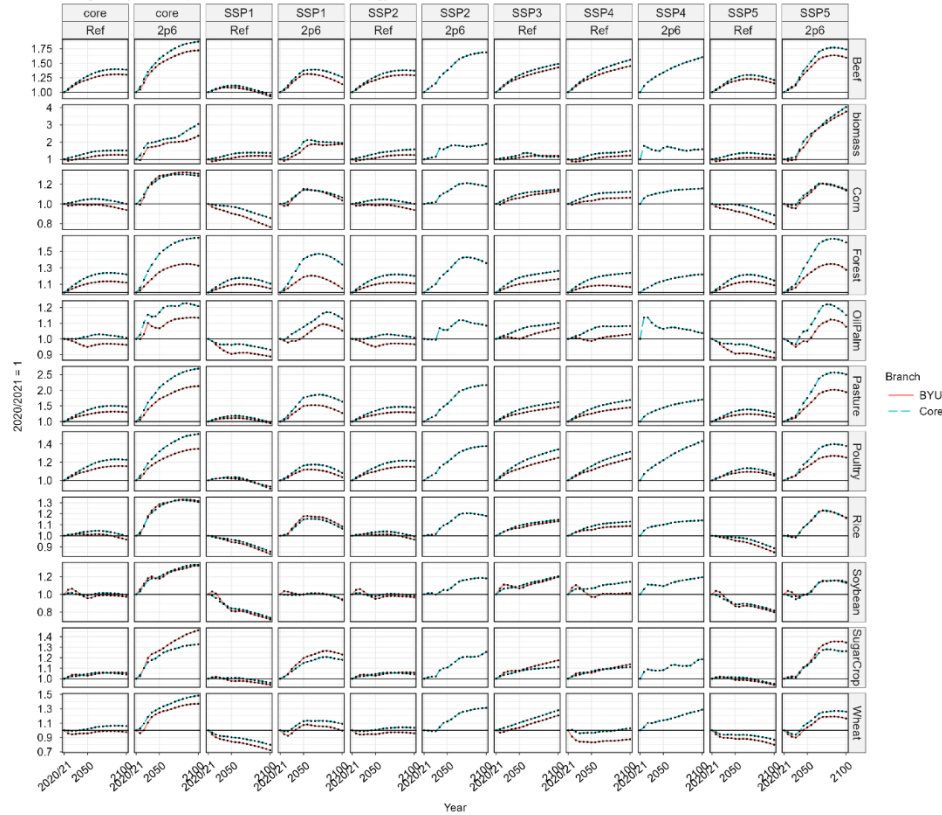
China: producer prices for key sectors

BYU generally has lower price increase due to lower 1G bioenergy

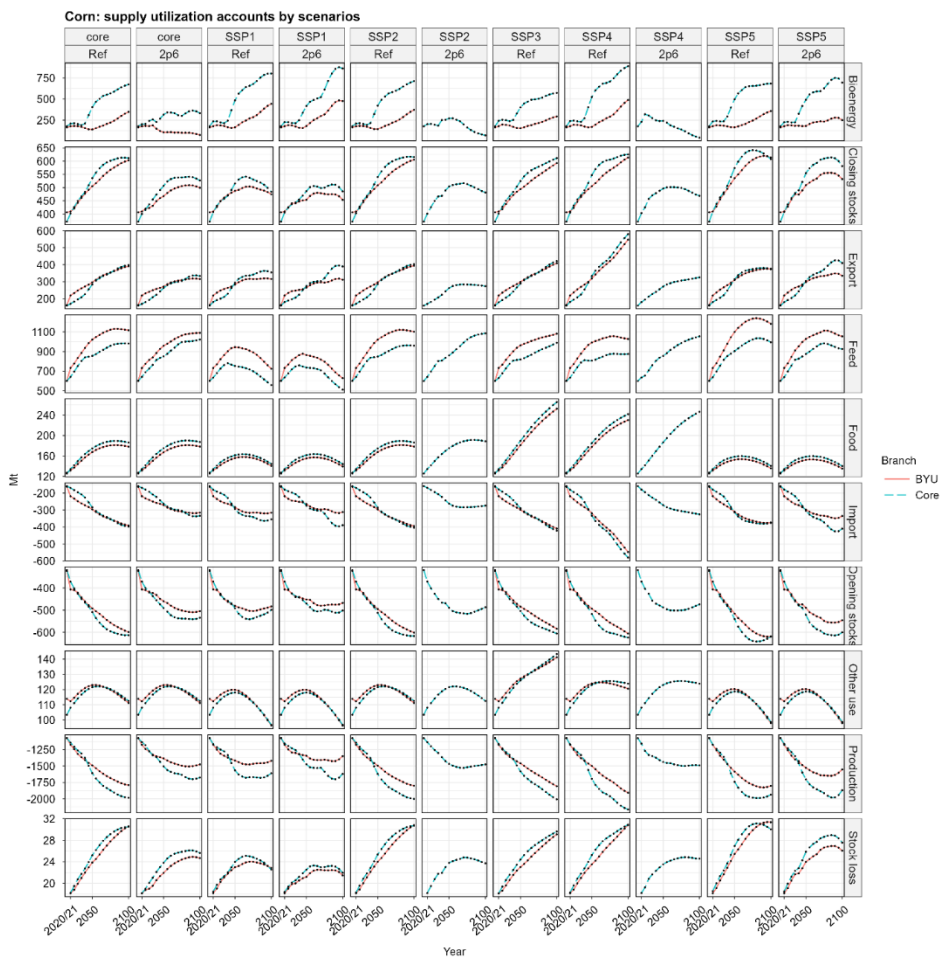
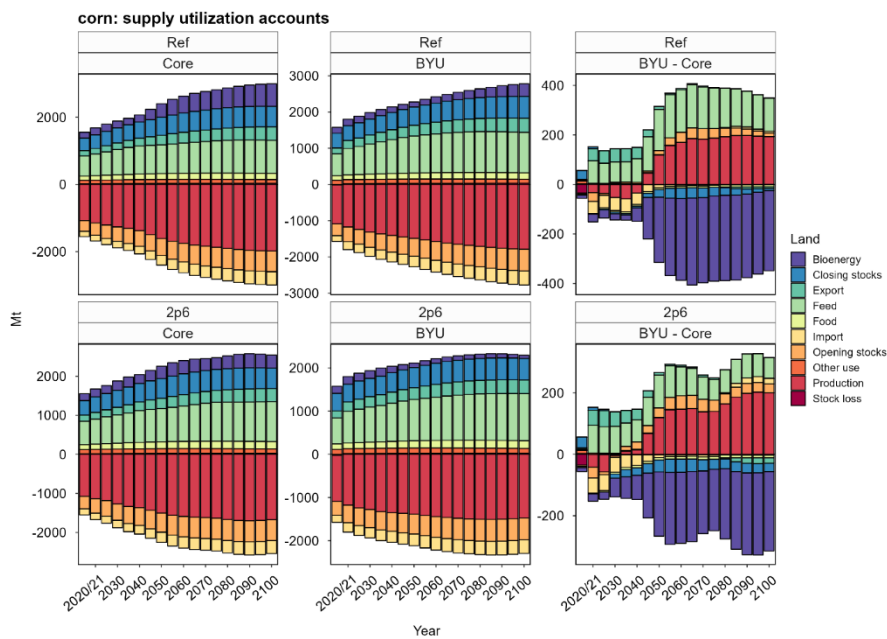


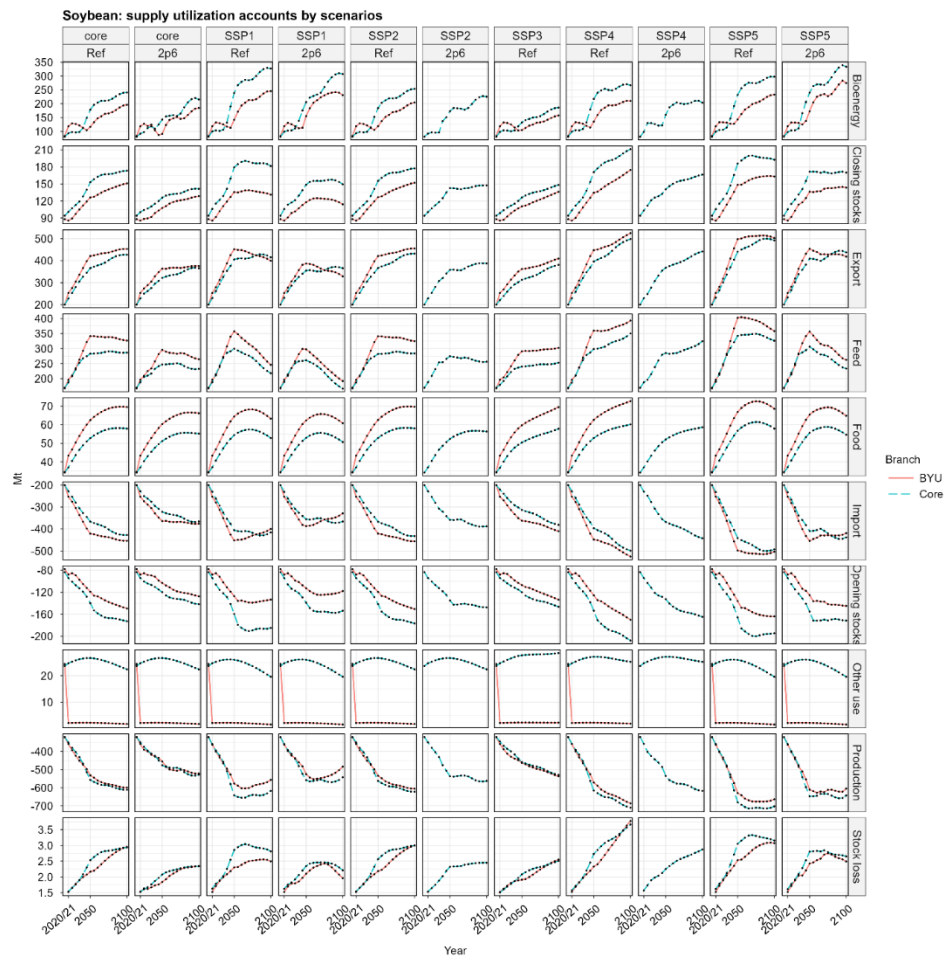
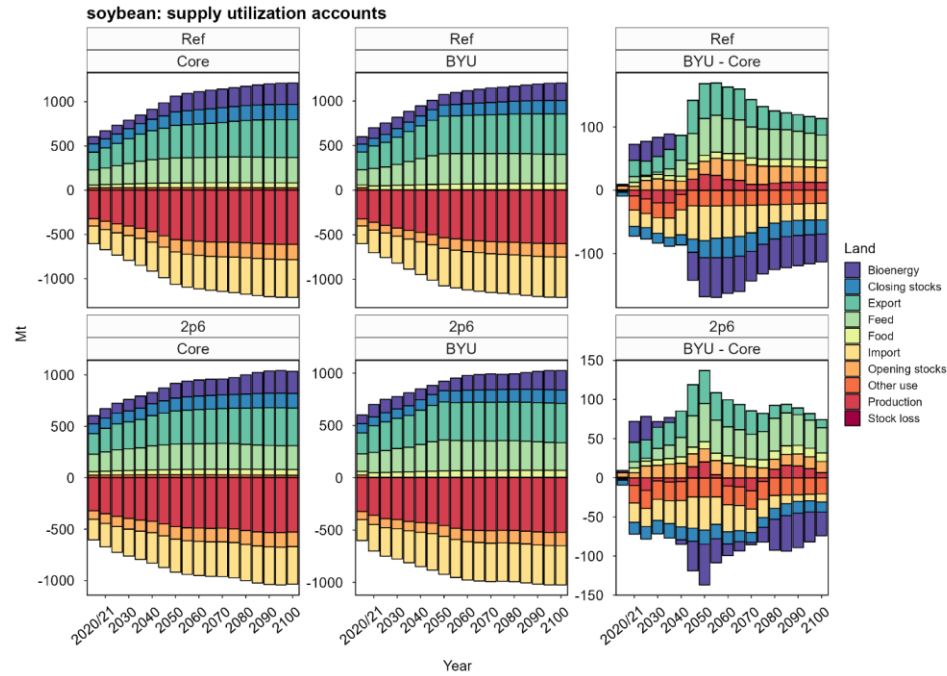
Indonesia: producer prices for key sectors

BYU generally has lower price increase due to lower 1G bioenergy

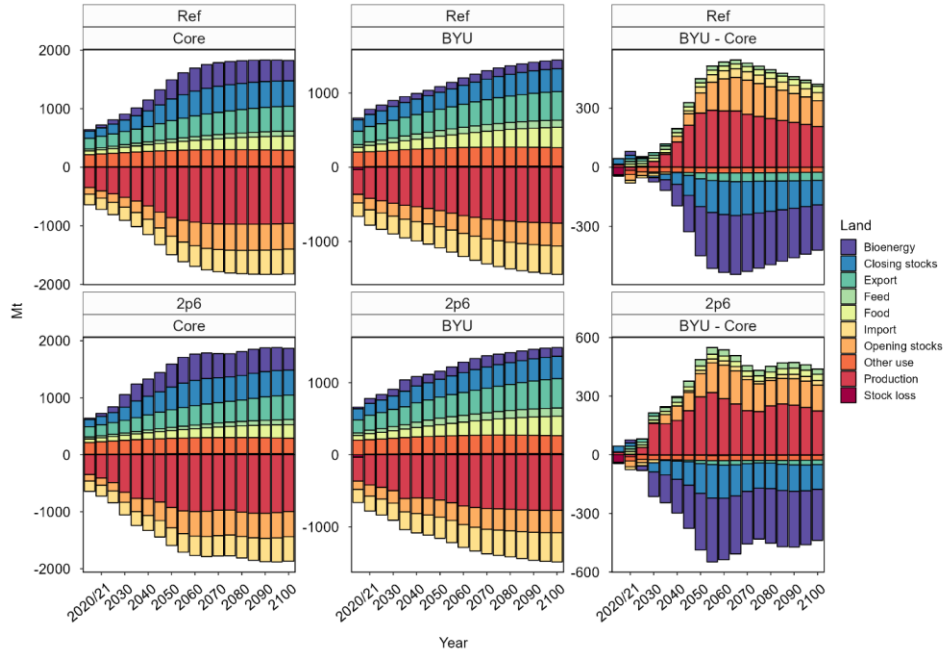


Supply utilization balance

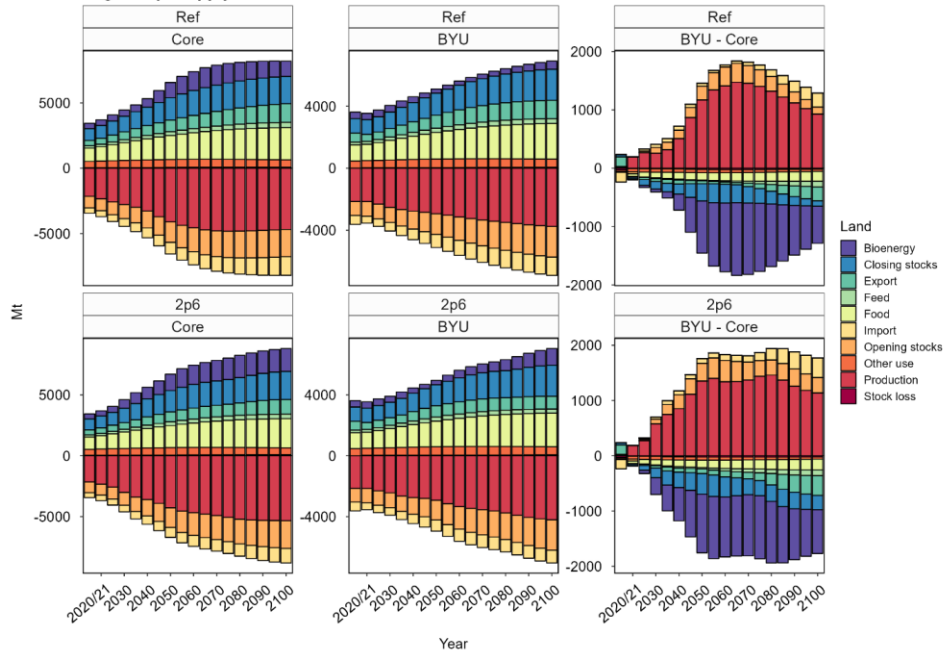


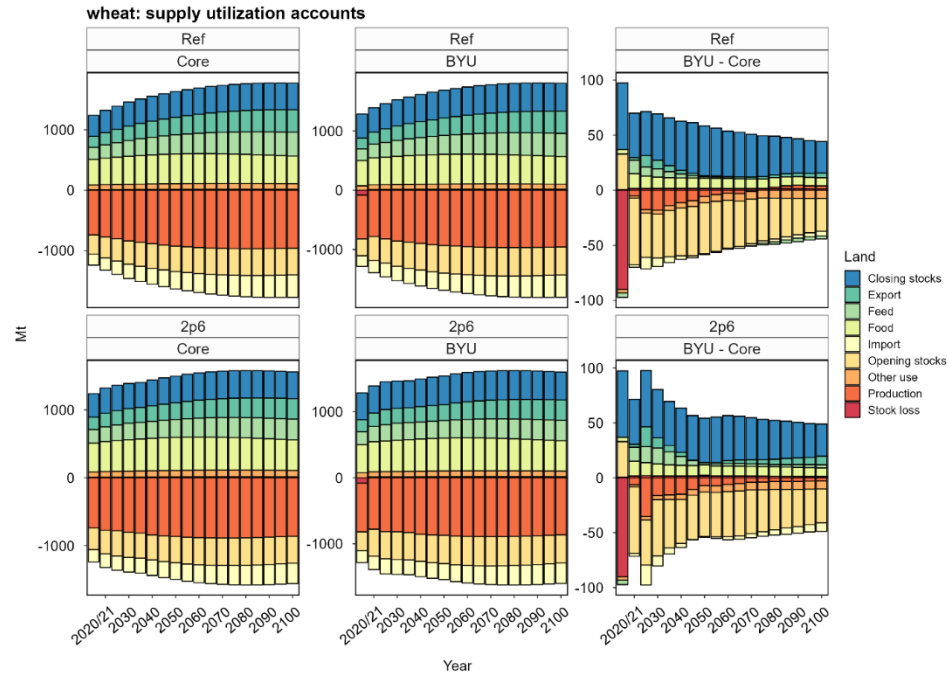


oilpalm: supply utilization accounts

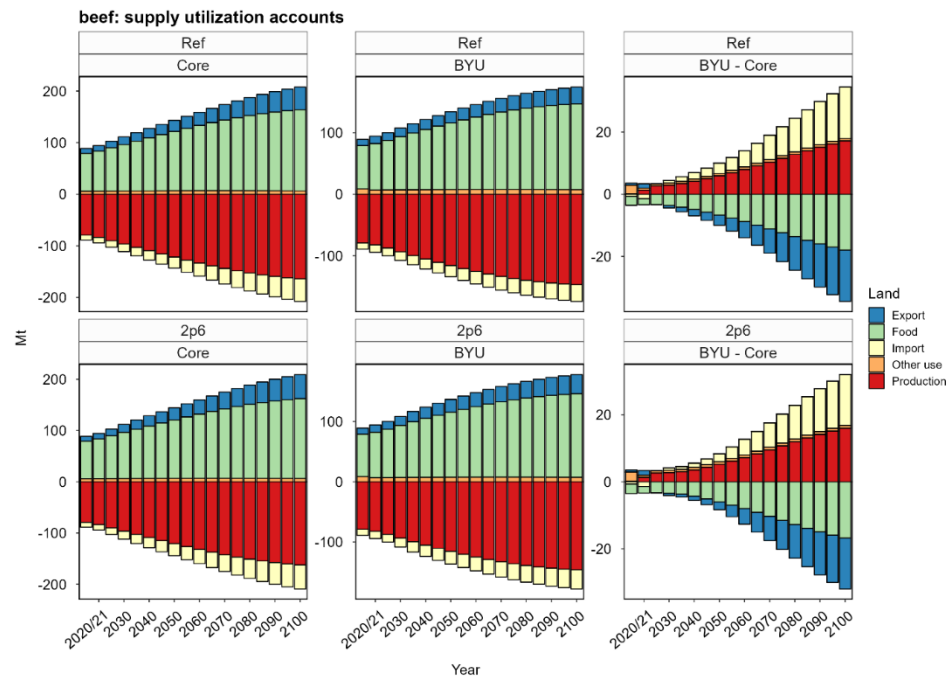


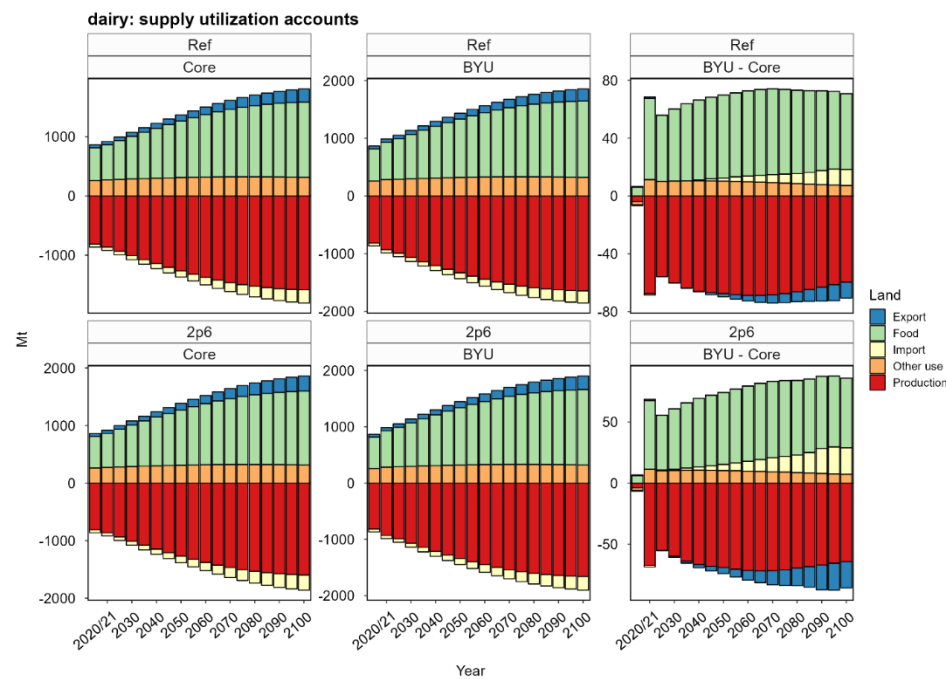
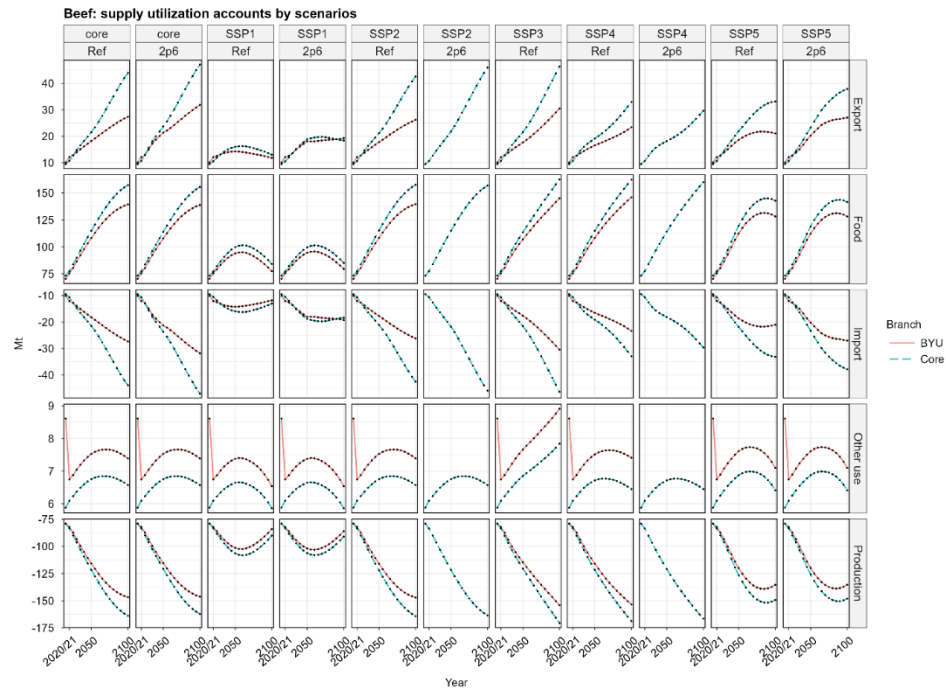
sugarcrop: supply utilization accounts

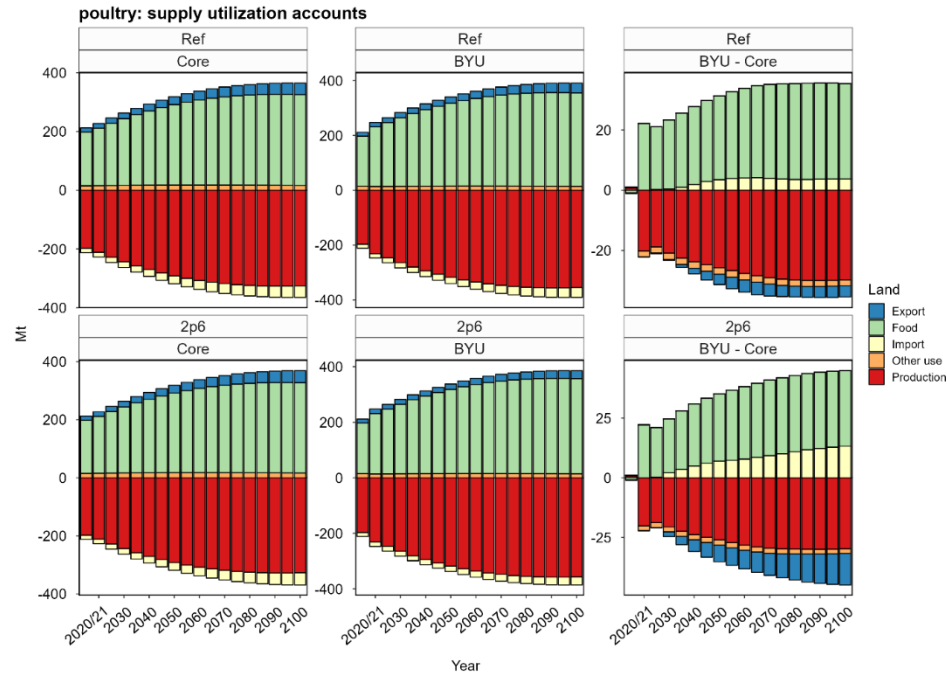




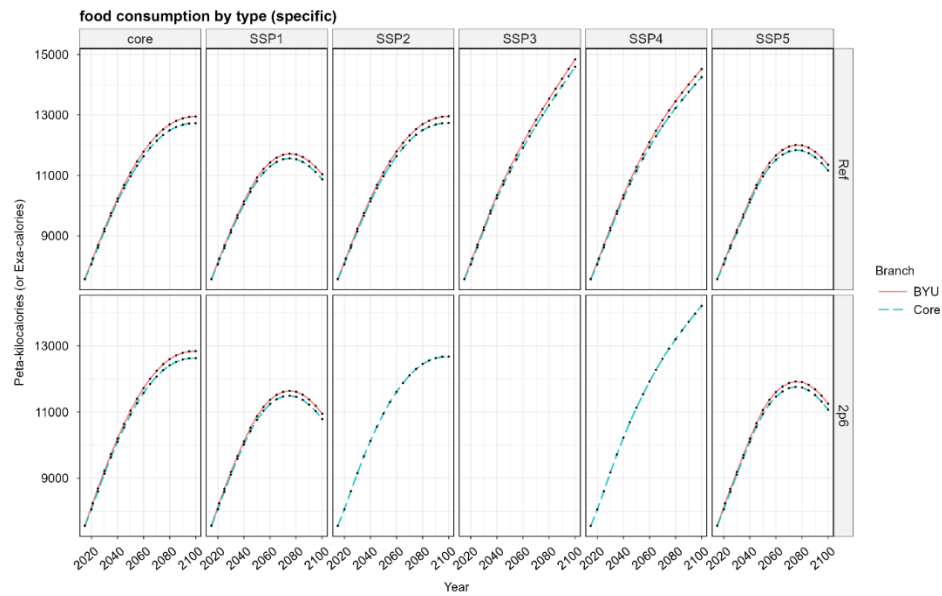
There will be relatively less food consumption from beef but more from dairy, poultry, and others.



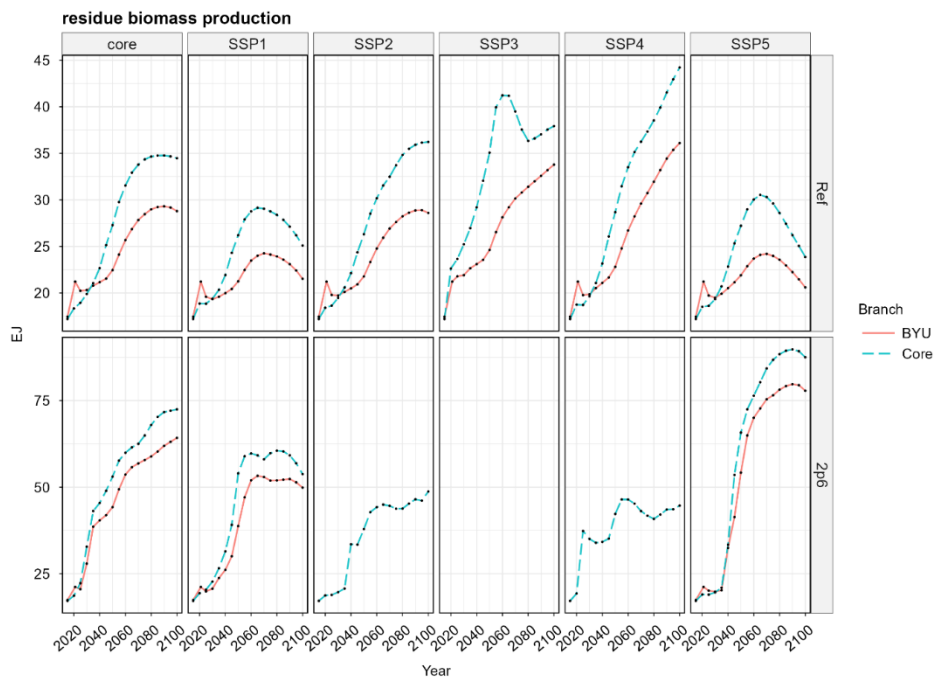
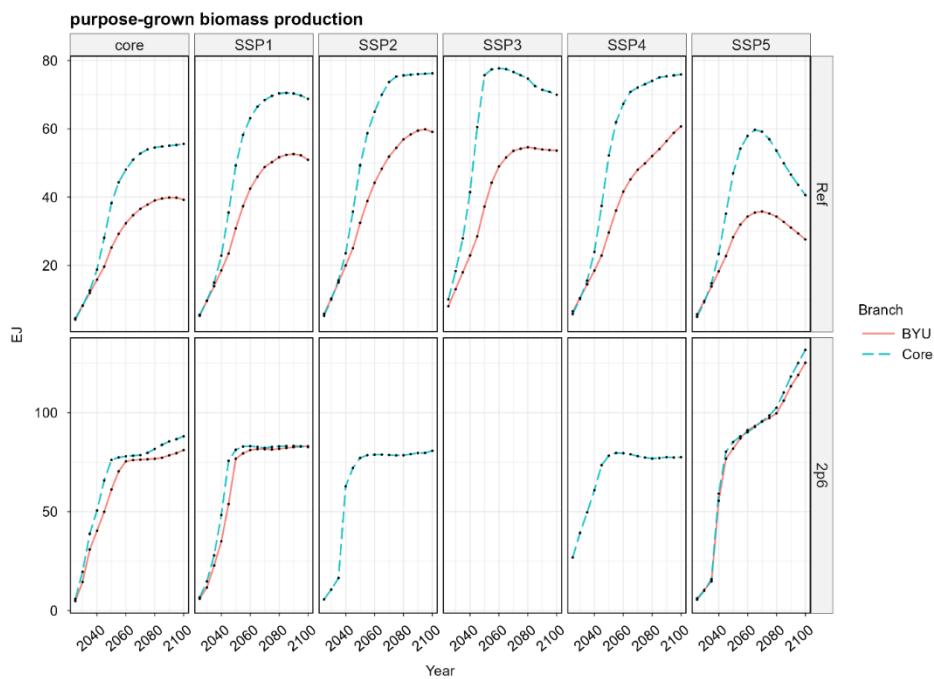




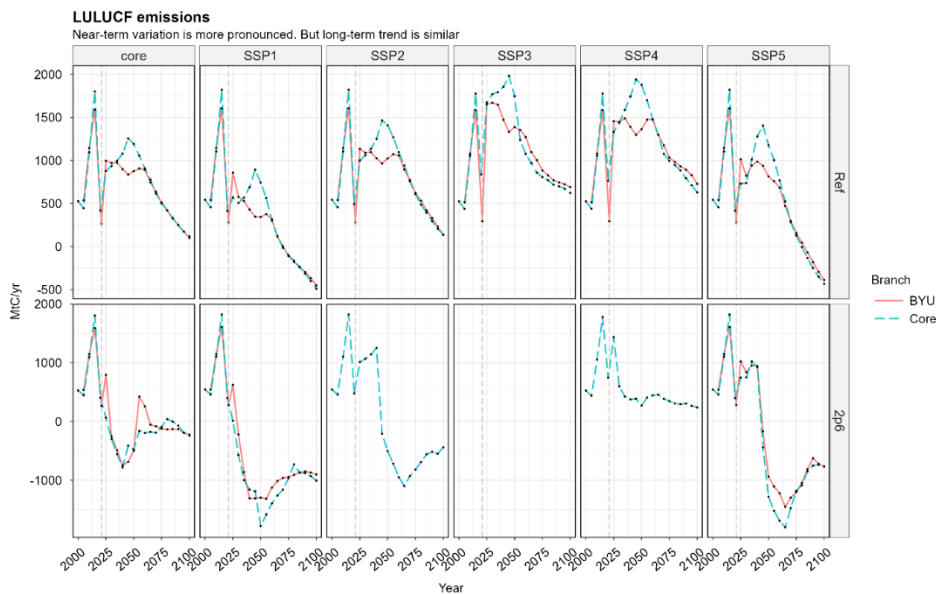
Food dietary energy supply



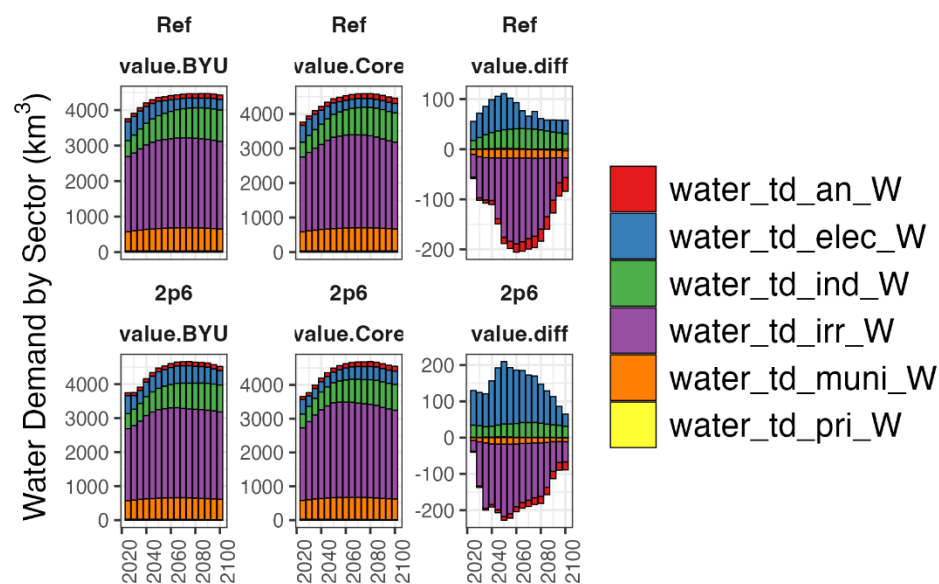
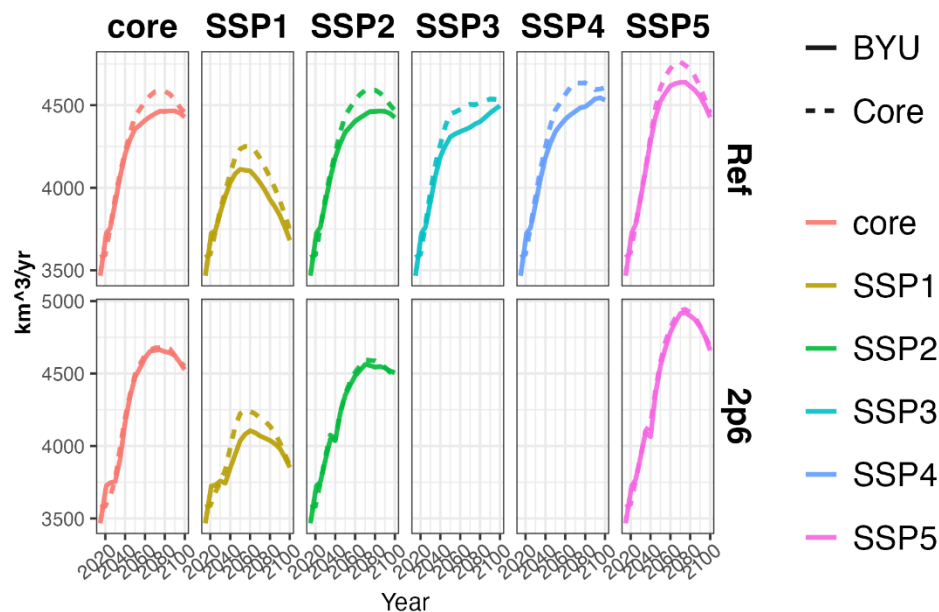
Purpose energy crop production and residual biomass supply



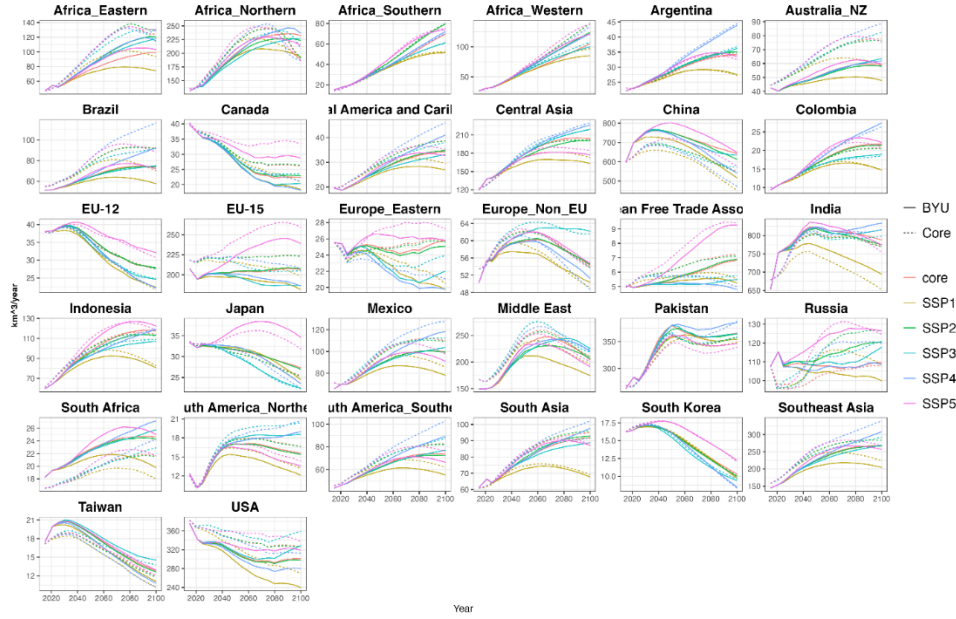
Land use change carbon emissions



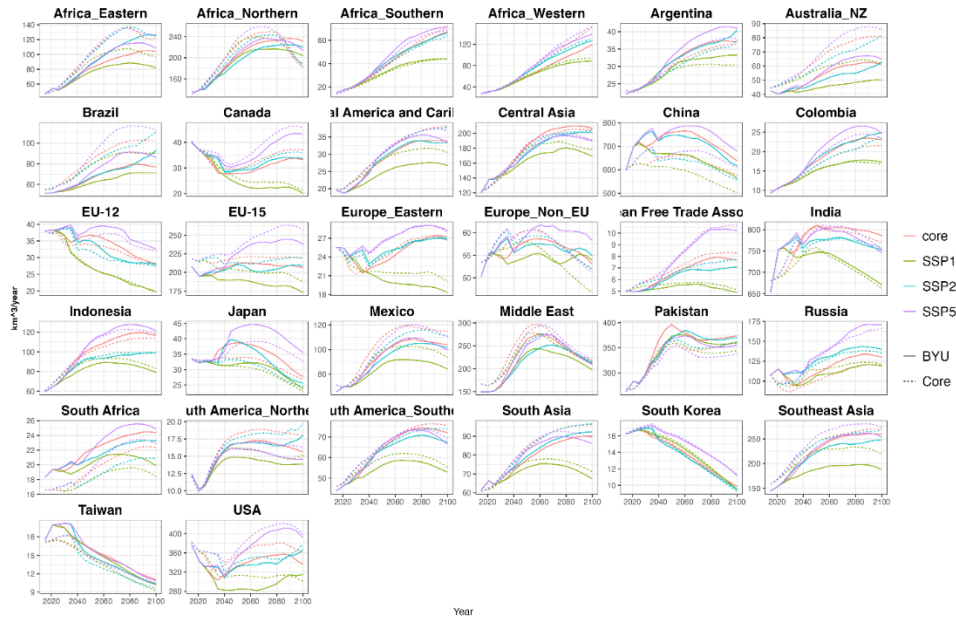
Water

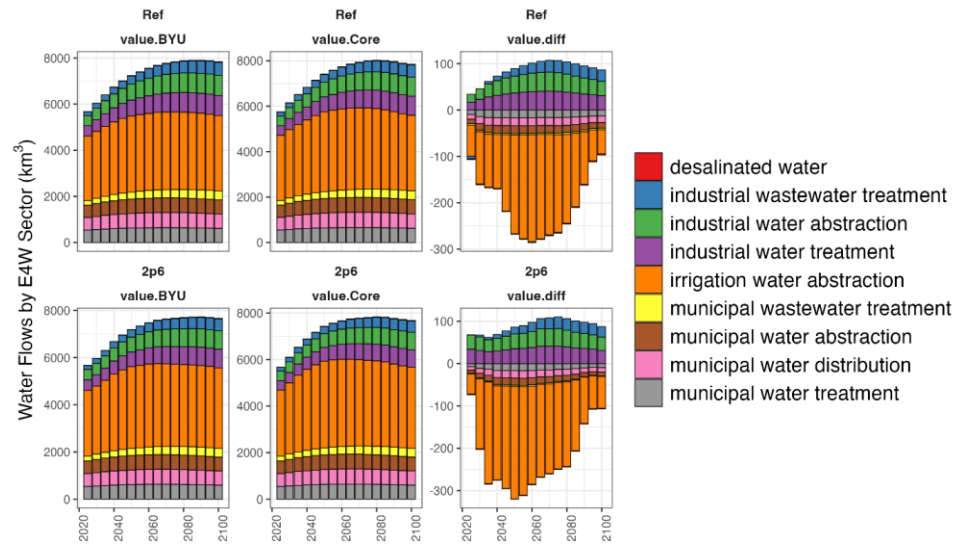


Regional Water Withdrawals - No Policy

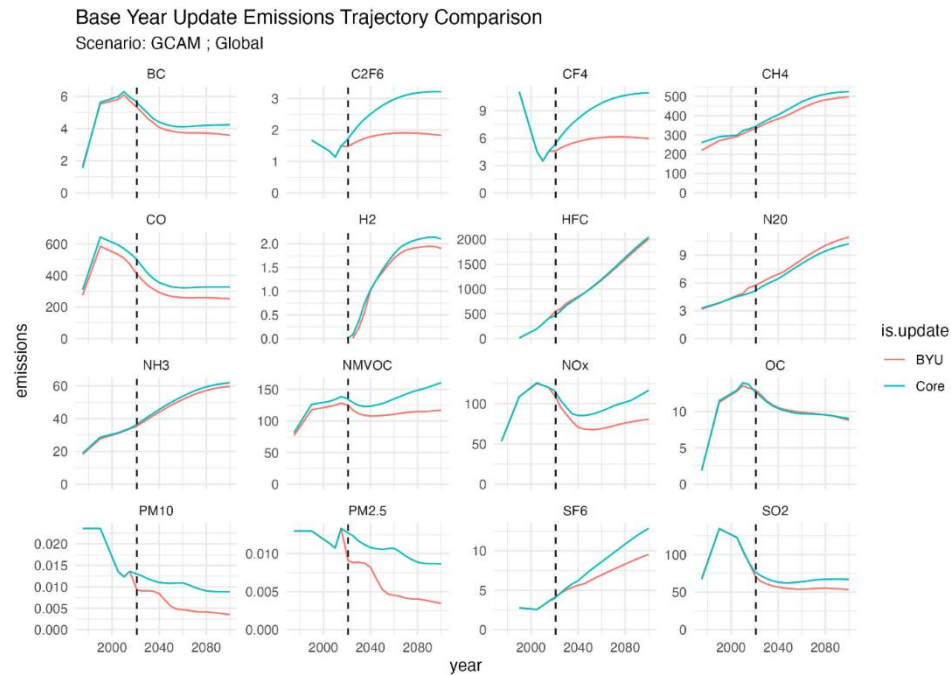


Regional Water Withdrawals - 2p6



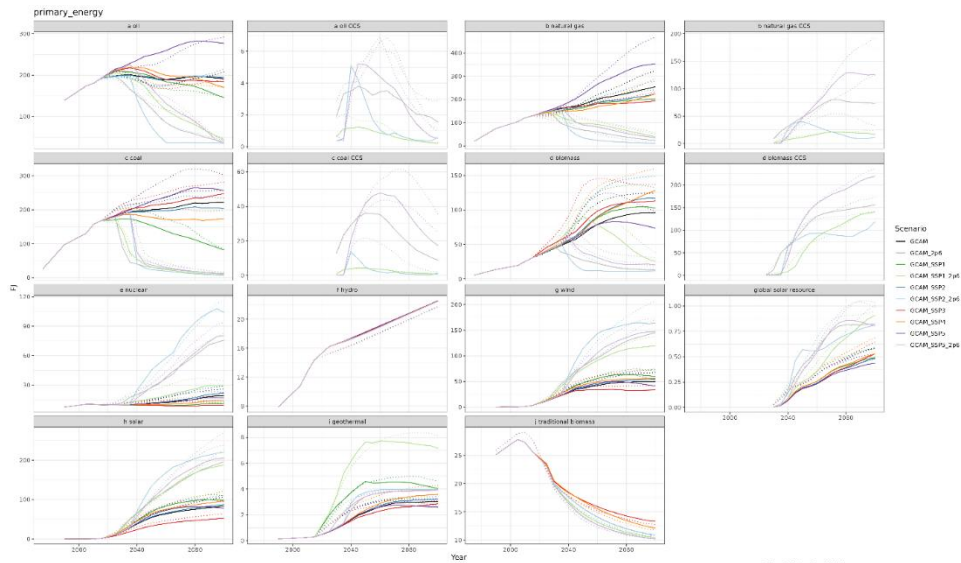


Emissions

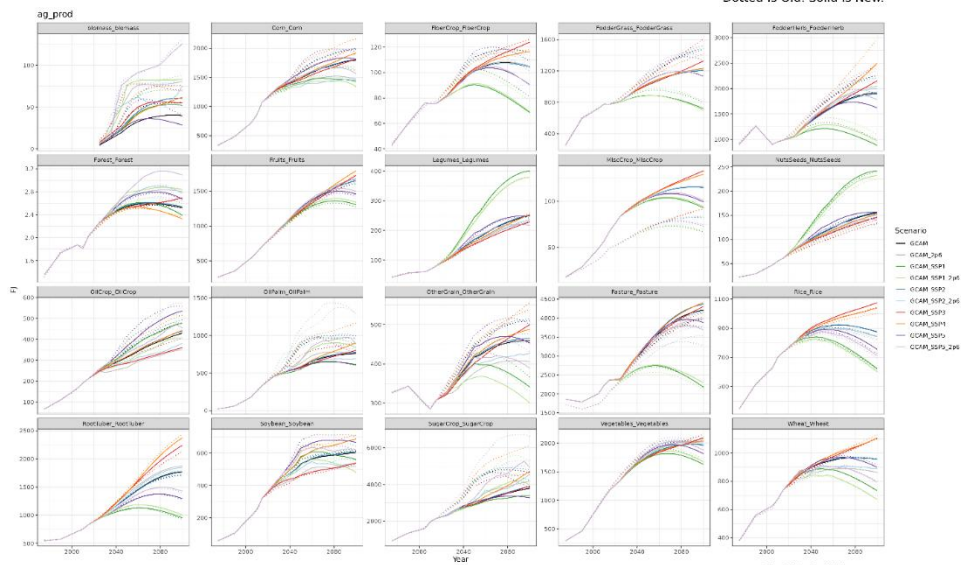


Emissions are generally lower across the board, which is a combination of lower emissions in the new 2021 base year (dotted vertical line) and lower drivers (energy consumption and service demands) into the future. In particular, lower "other industrial" sector service demand directly translates to lower industrial process emissions (for example C2F6 and part of SF6).

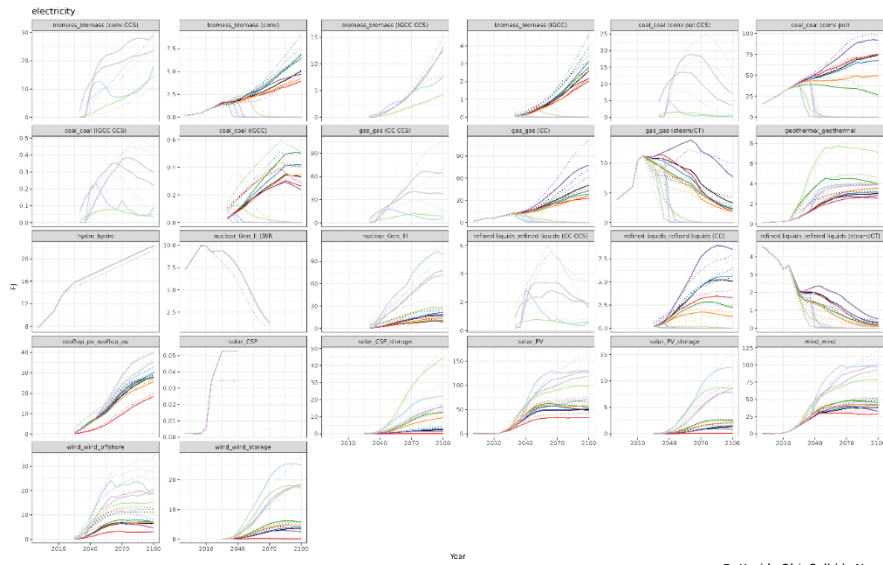
Anomaly Detector Figures



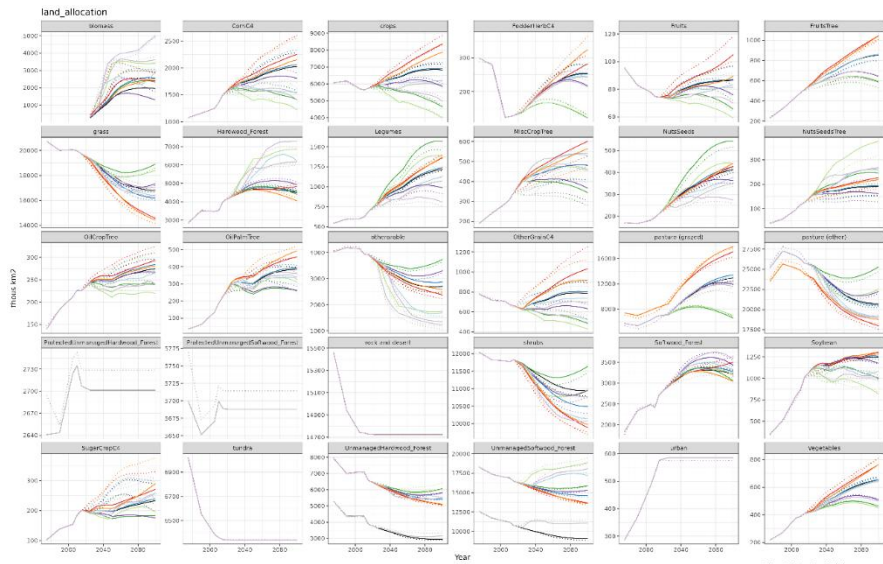
Dotted is Old. Solid is New.



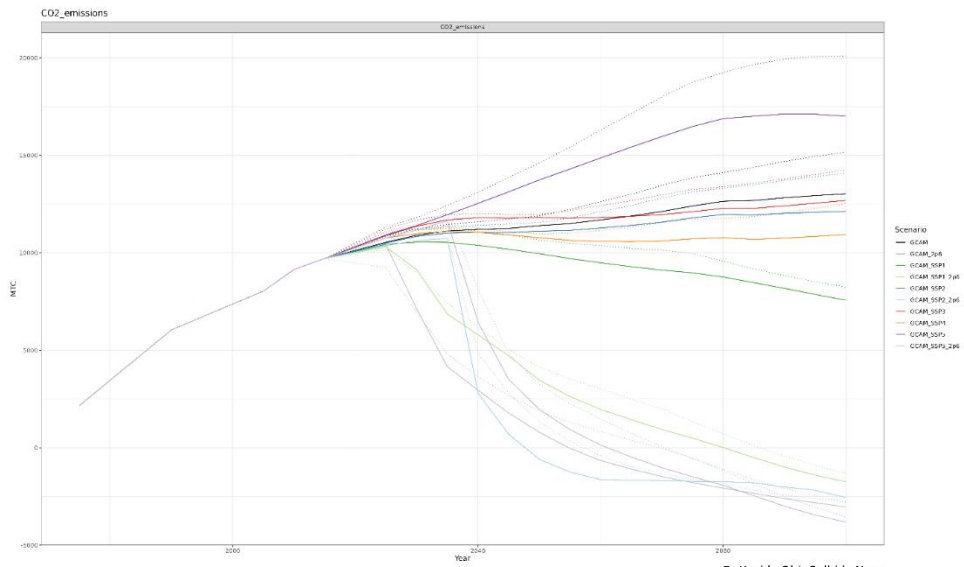
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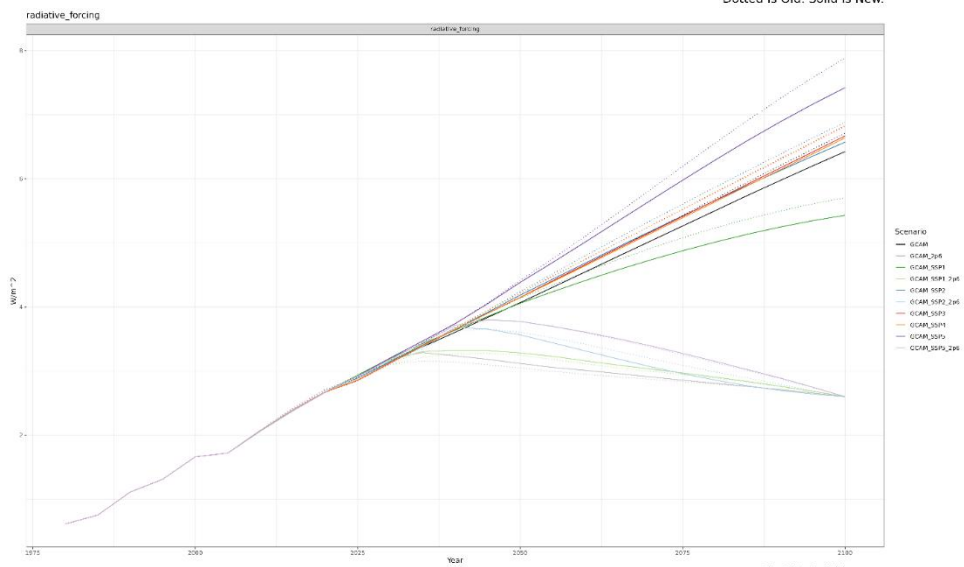
Dotted is Old. Solid is New.



Dotted is Old. Solid is New.



Dotted is Old. Solid is New.



Dotted is Old. Solid is New.

